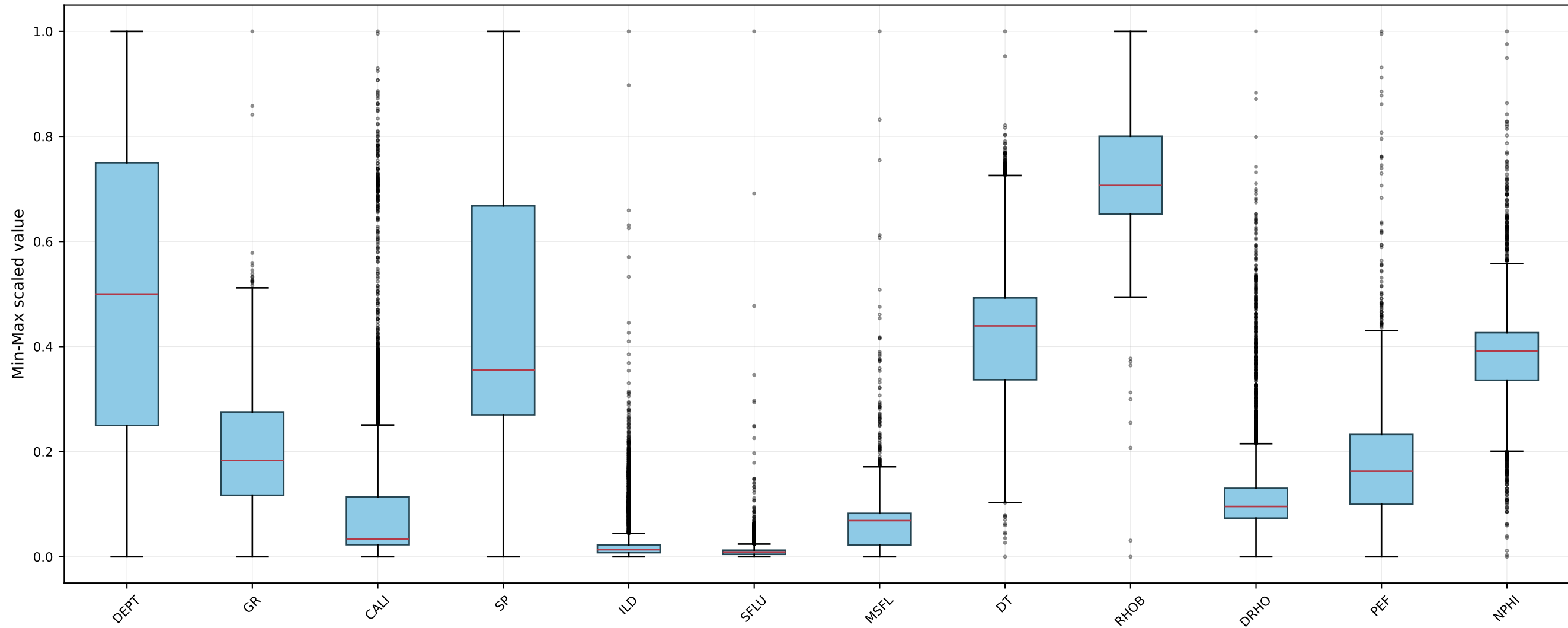
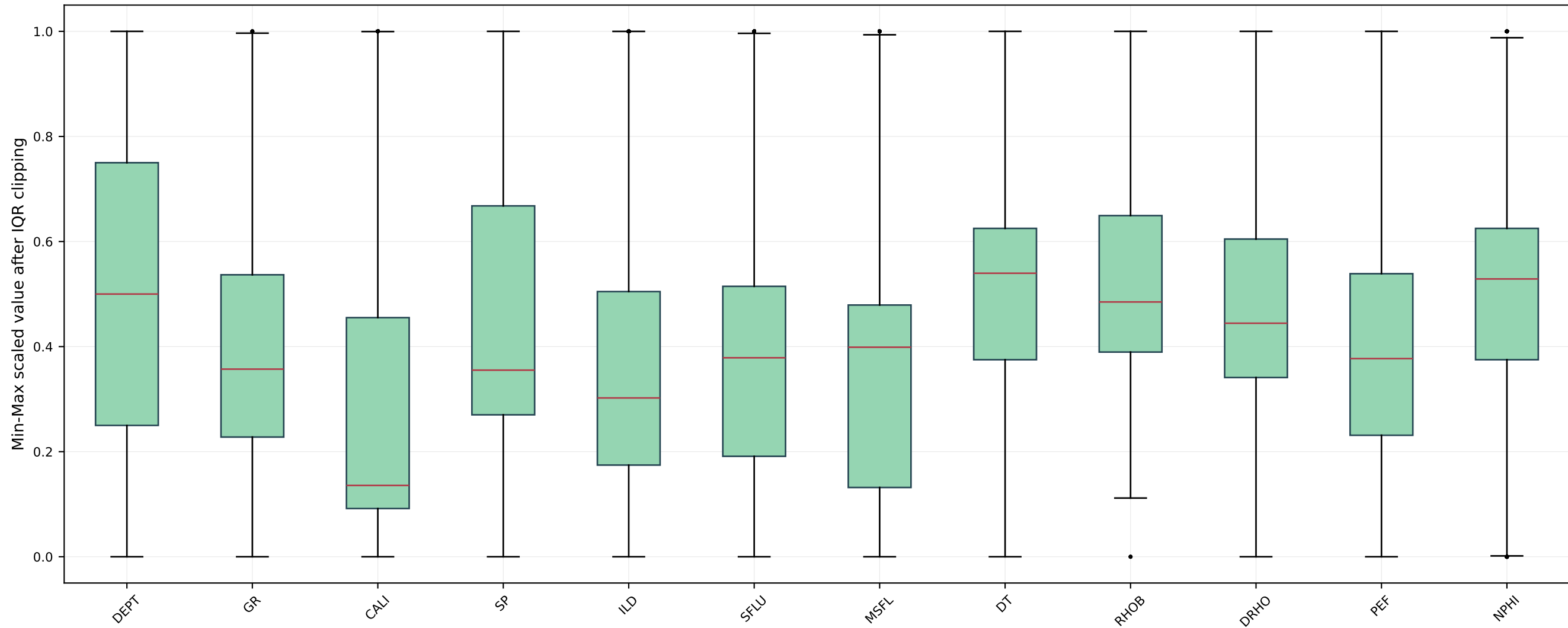


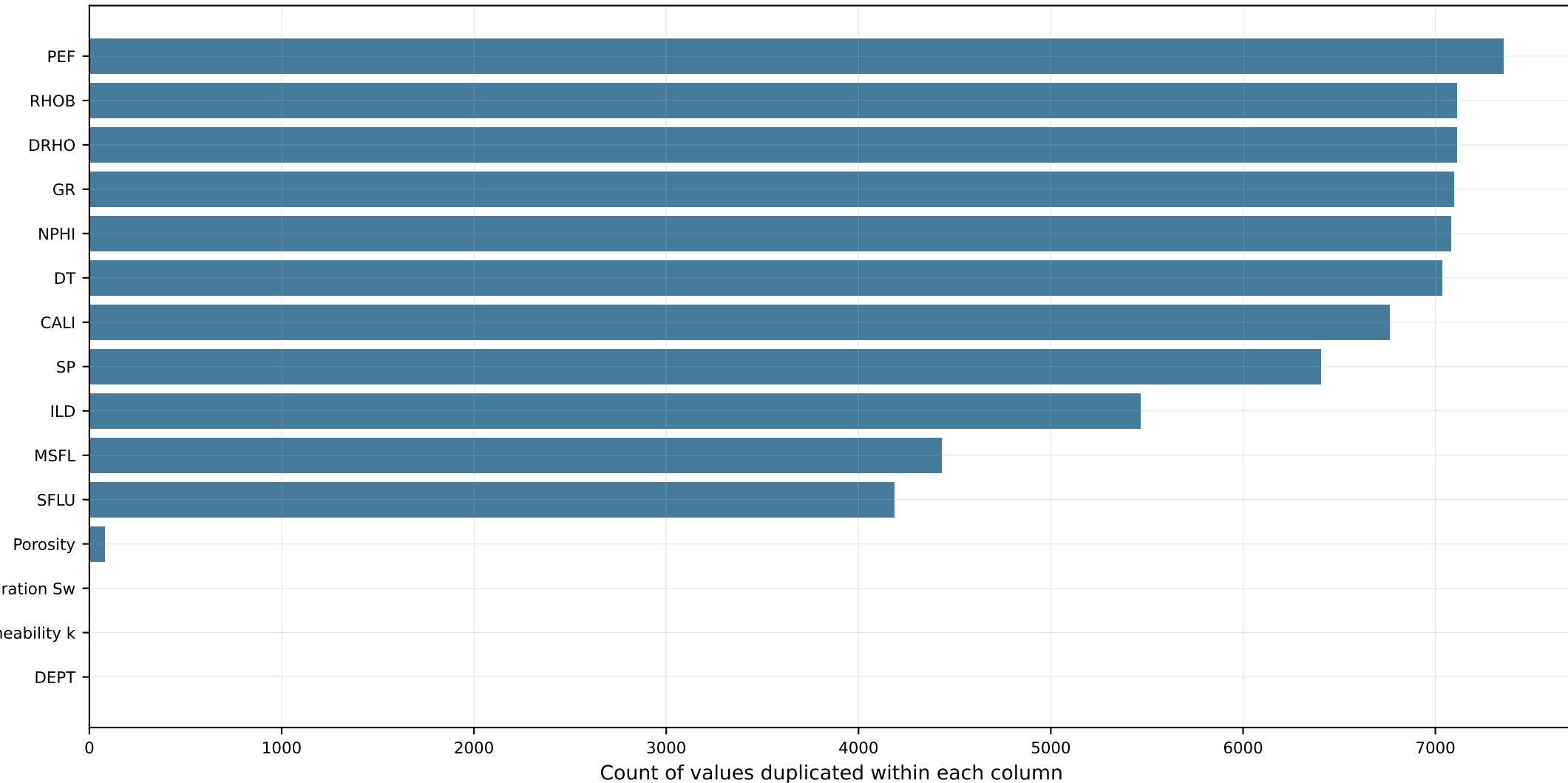
Boxplot for Input-Feature Outliers Before IQR Treatment



Boxplot for Input-Feature Outliers After IQR Treatment

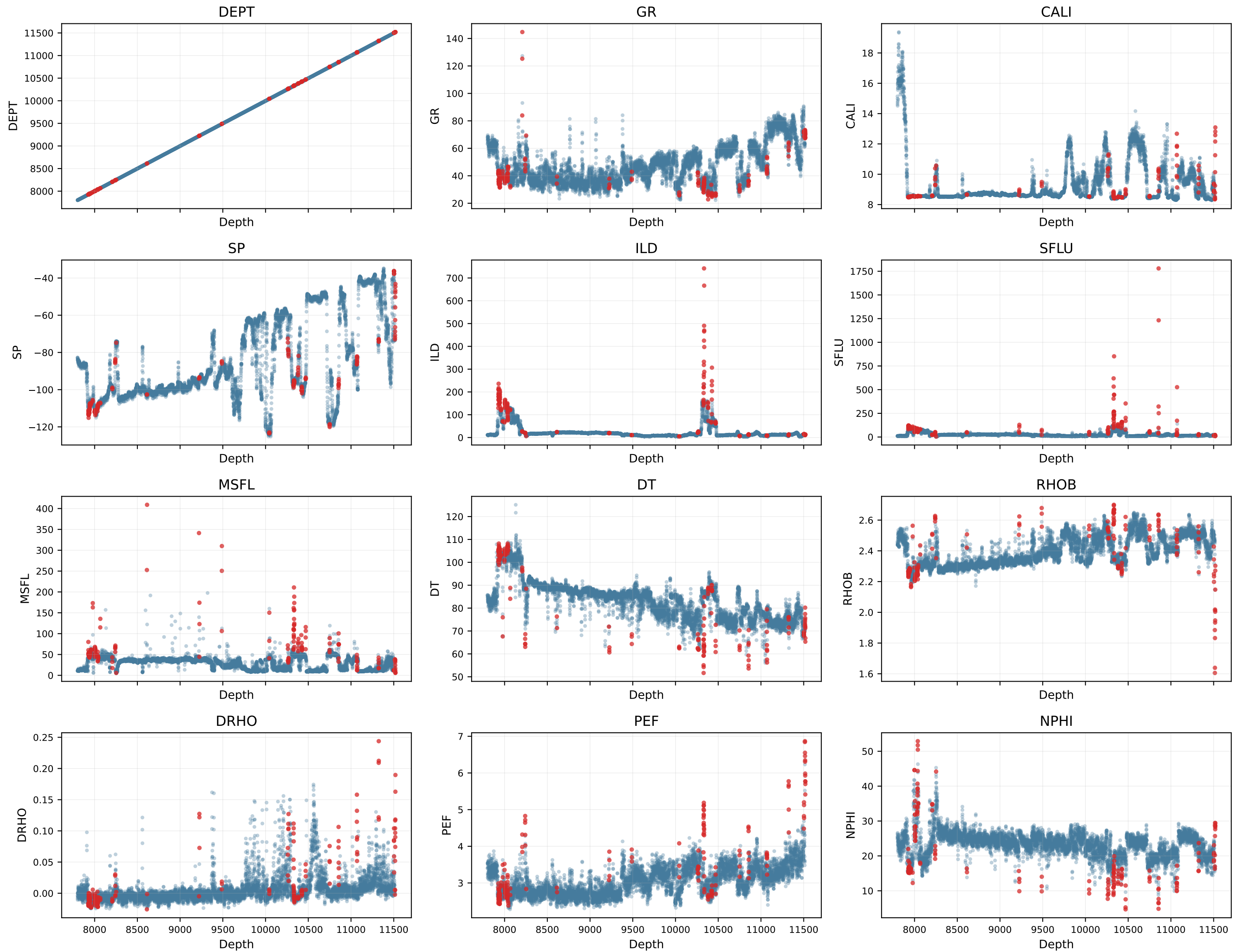


Histogram for Duplicate Data by Column (full-row duplicates = 0)



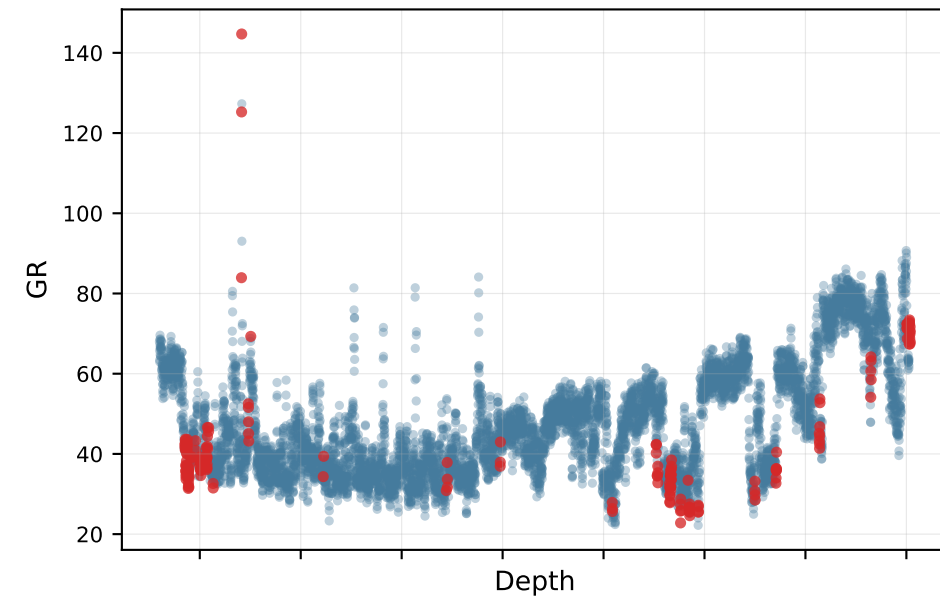
Scattered Plot for Noisy Data Detected by IsolationForest (n = 186)

Normal
Noisy

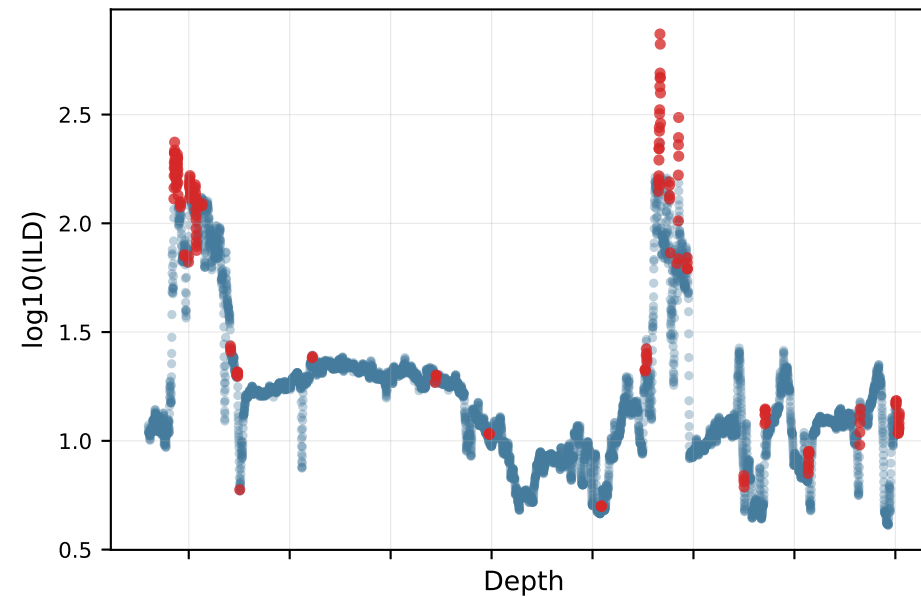


Scattered Plot for Noisy Data on Key Well-Log Curves

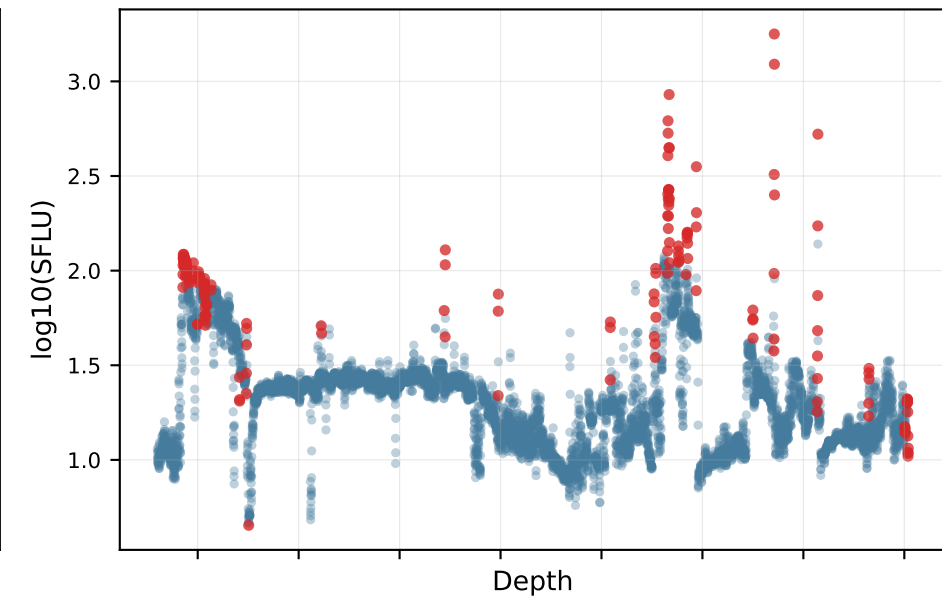
GR



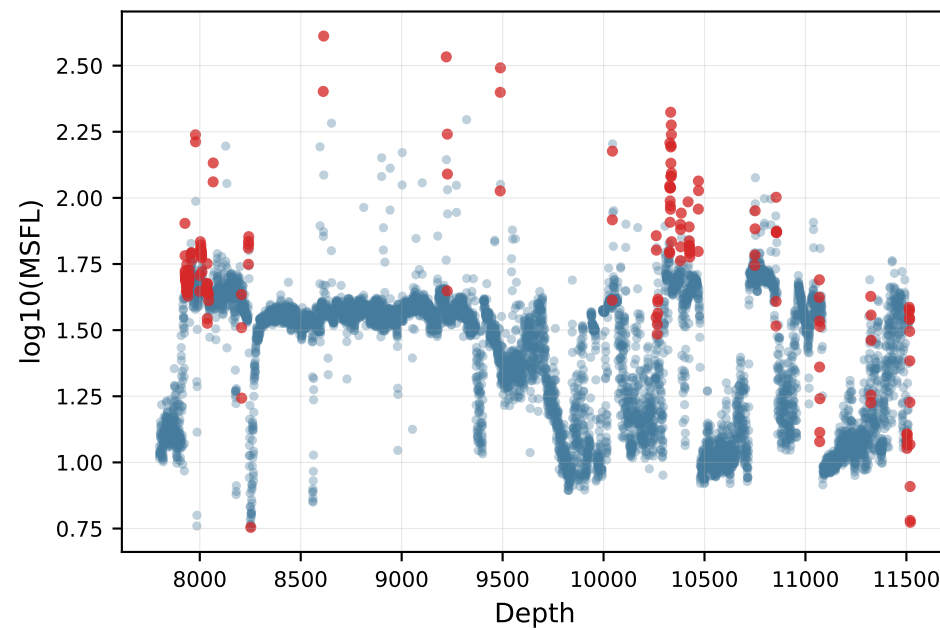
ILD



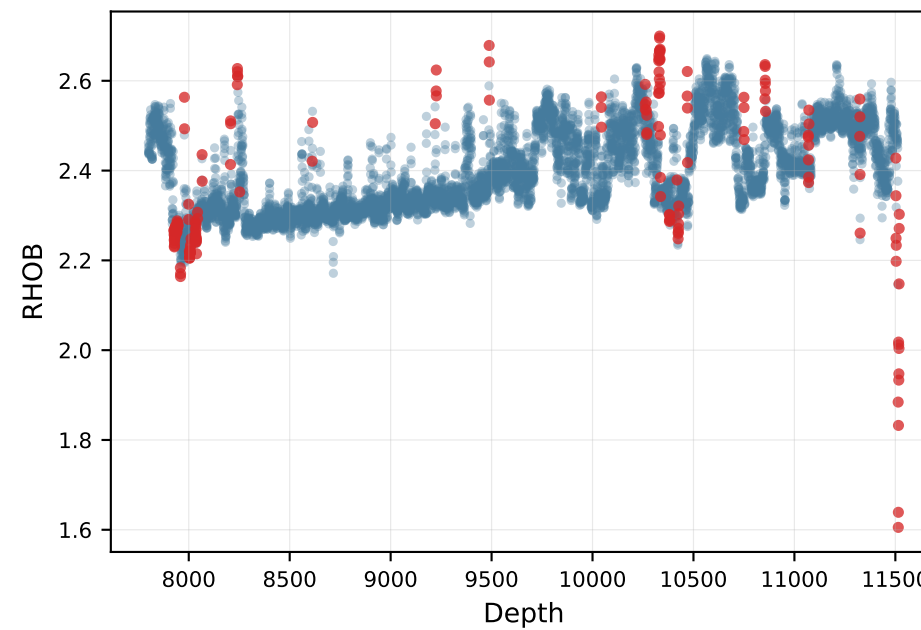
SFLU



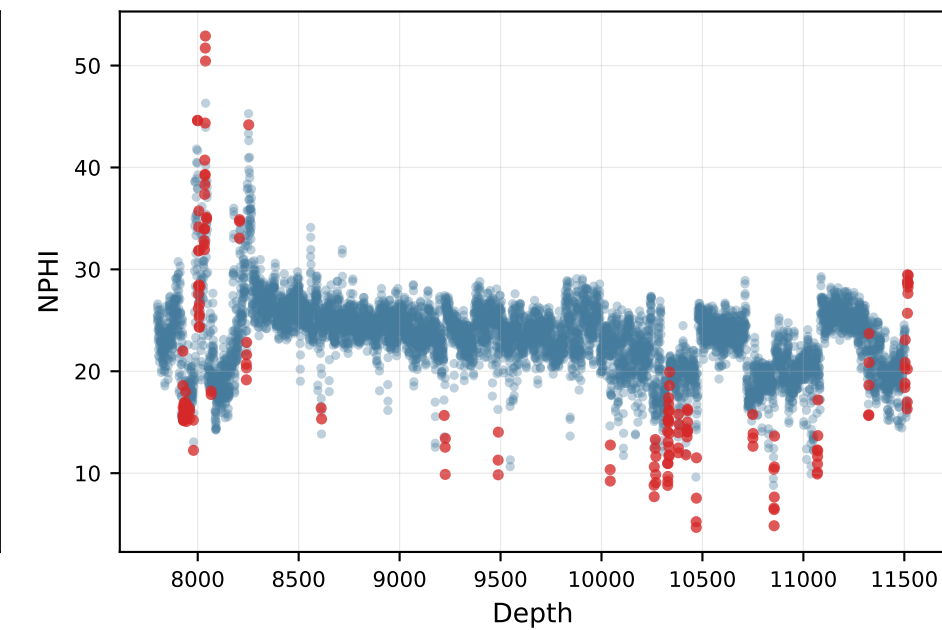
MSFL



RHOB

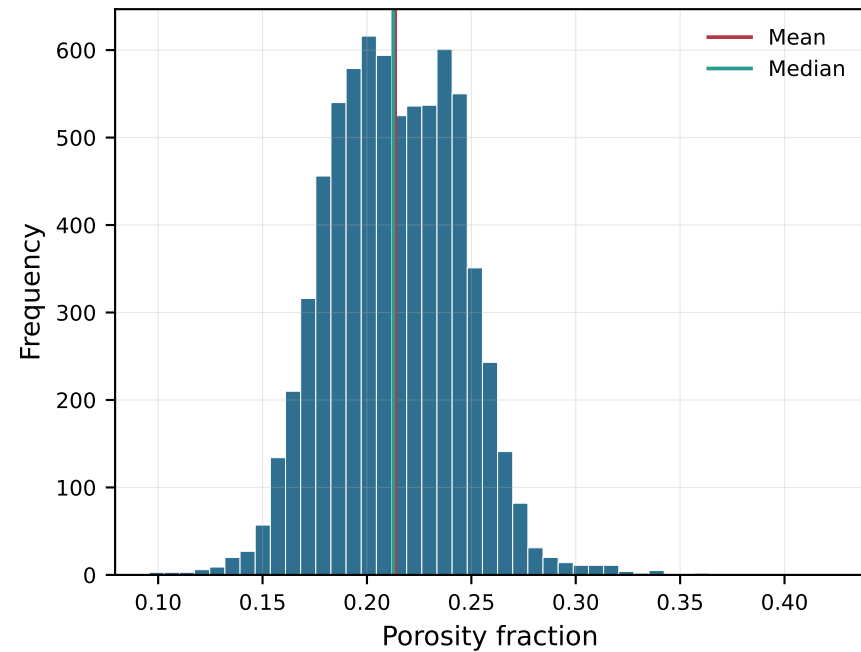


NPHI

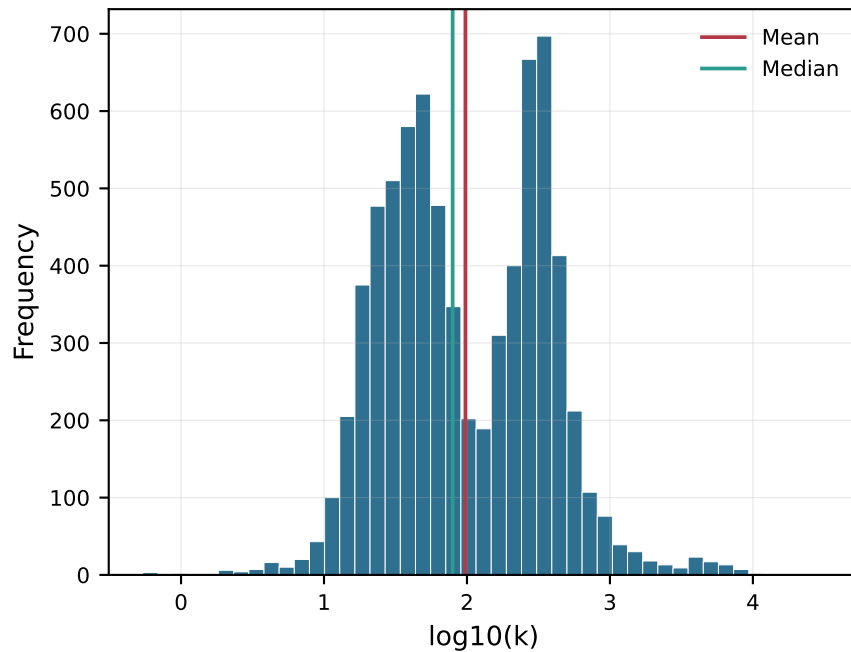


Distribution of Target Reservoir Properties

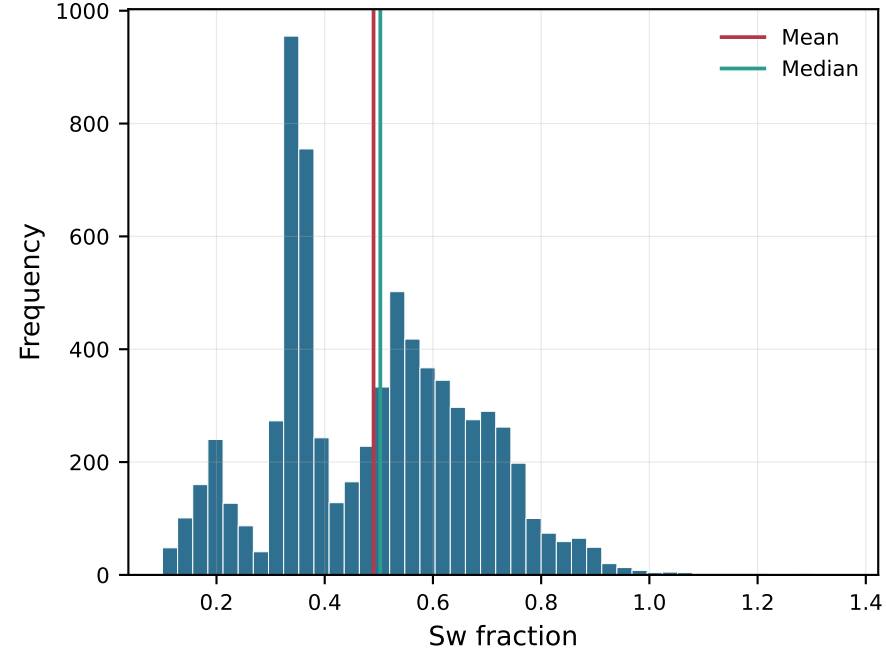
Porosity



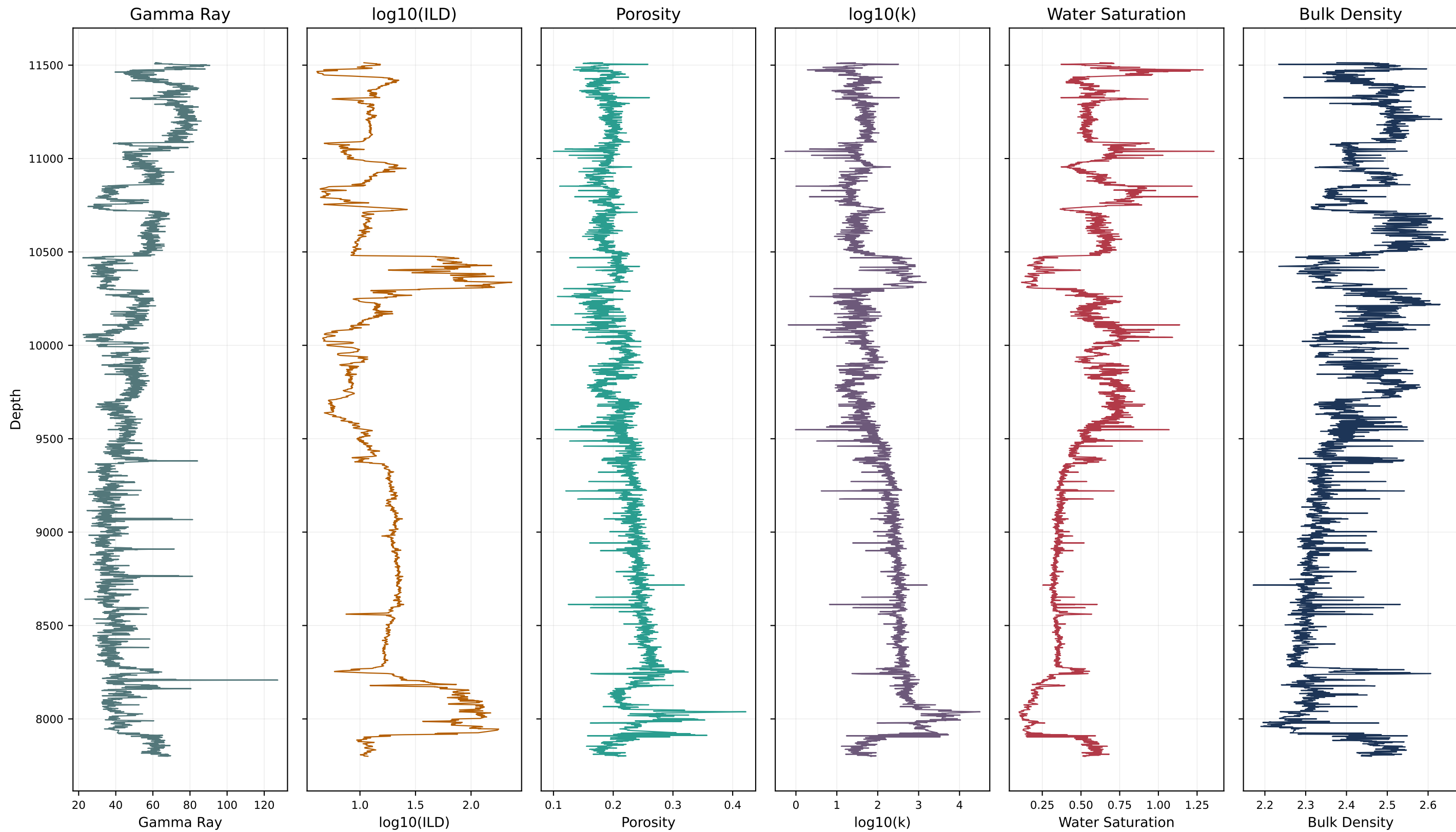
Permeability k



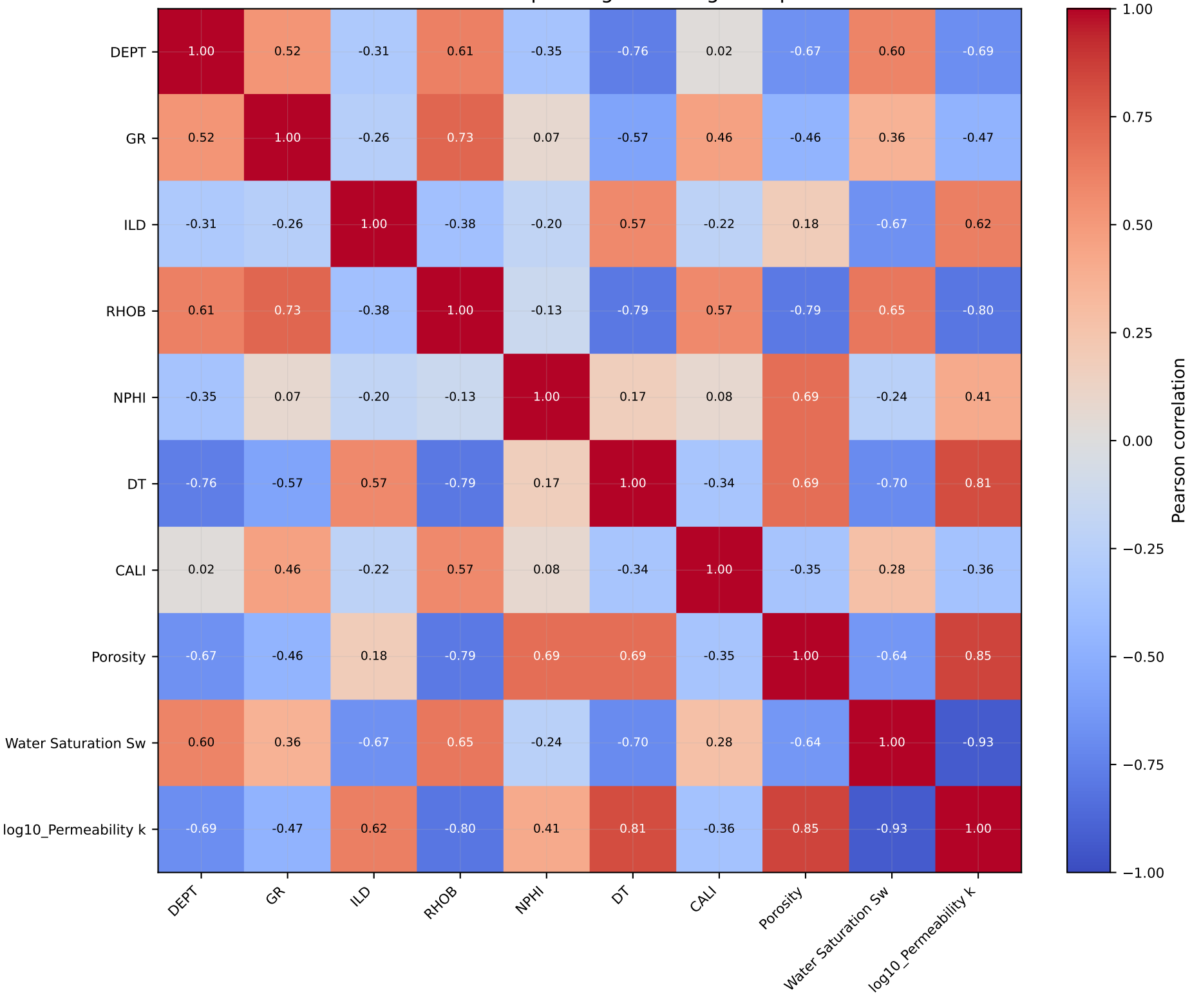
Water saturation



Well Log and Reservoir Property Profiles with Depth

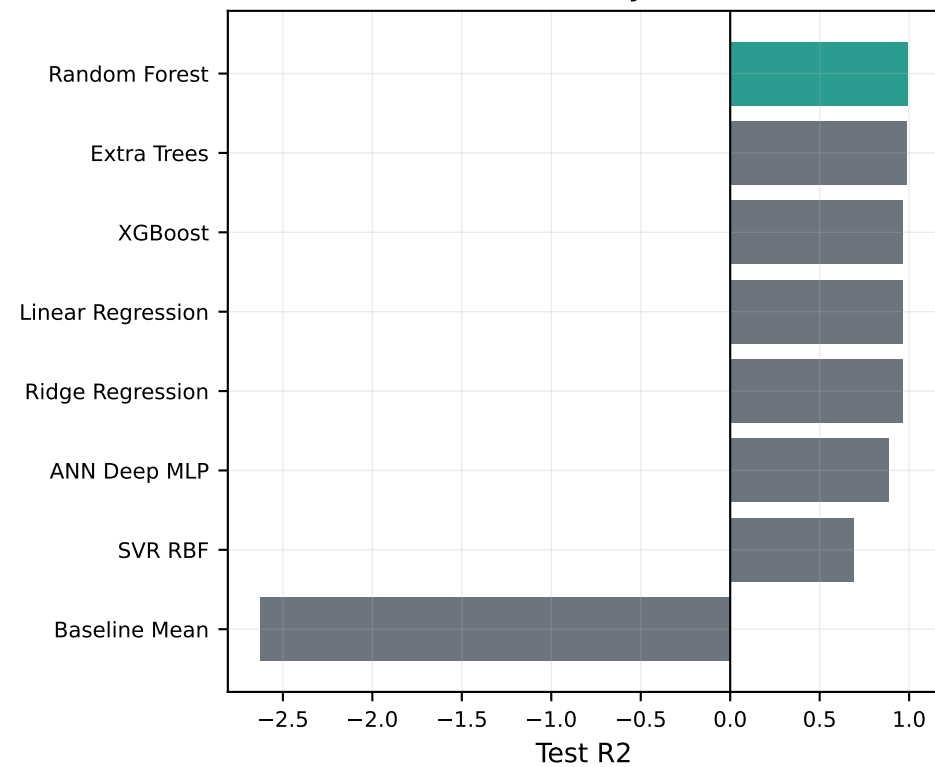


Correlation Heatmap of Logs and Target Properties

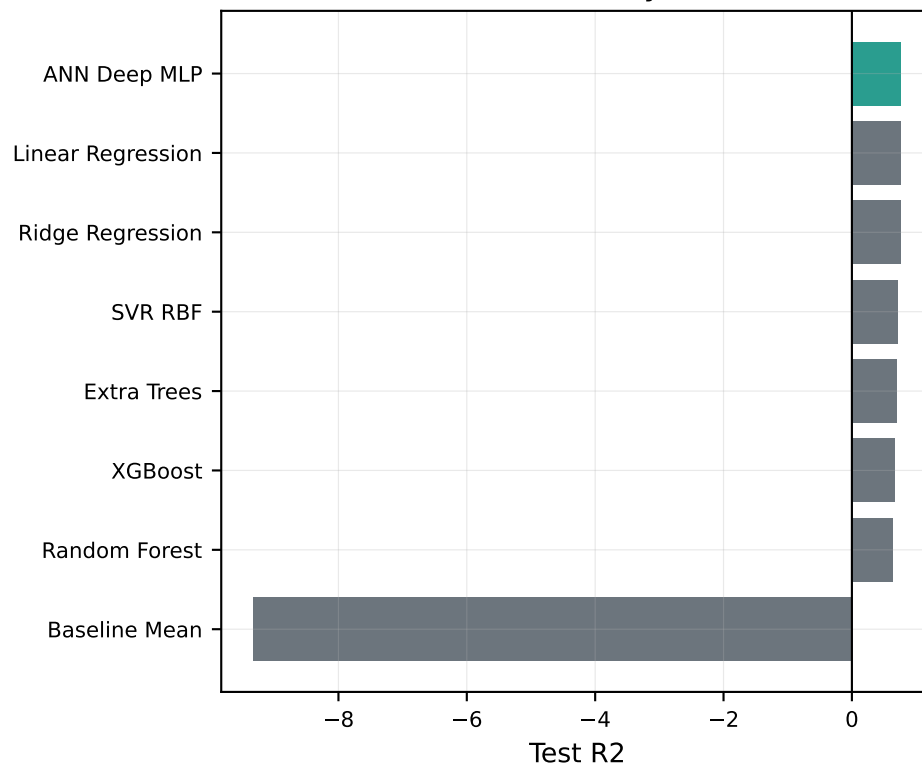


Model Comparison Using Depth-Holdout Test R2

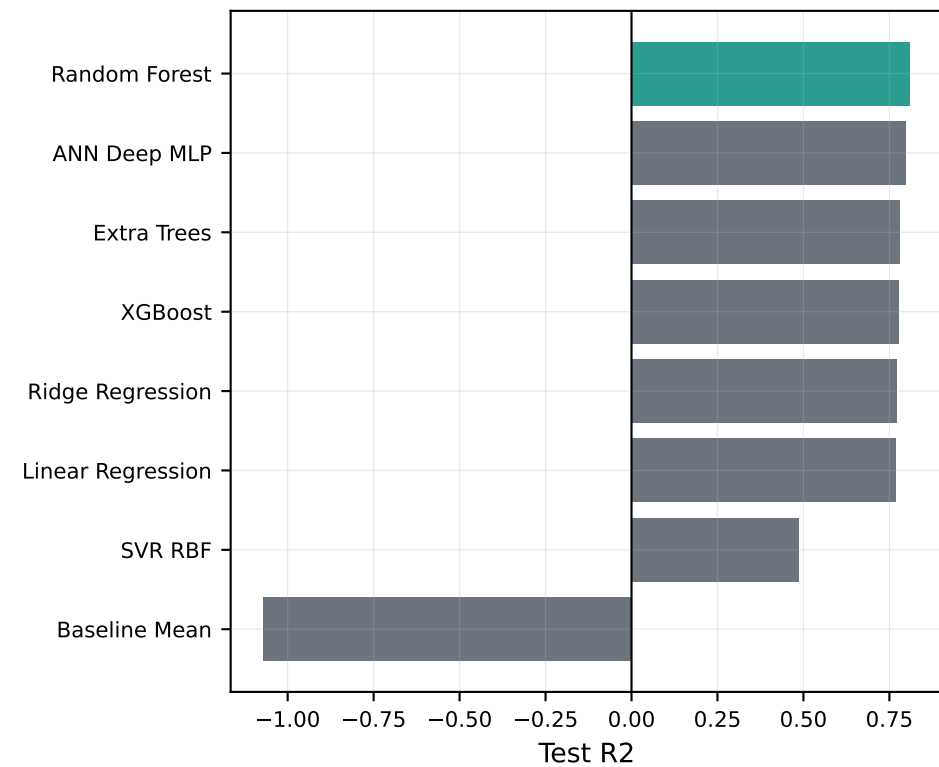
Porosity



Permeability k

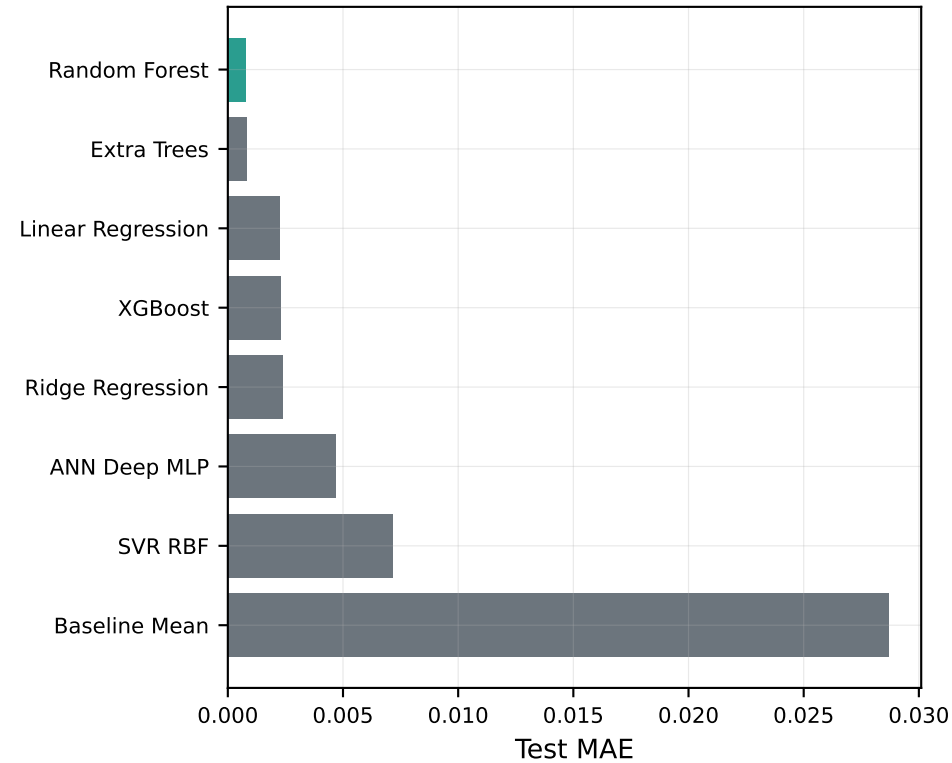


Water Saturation Sw

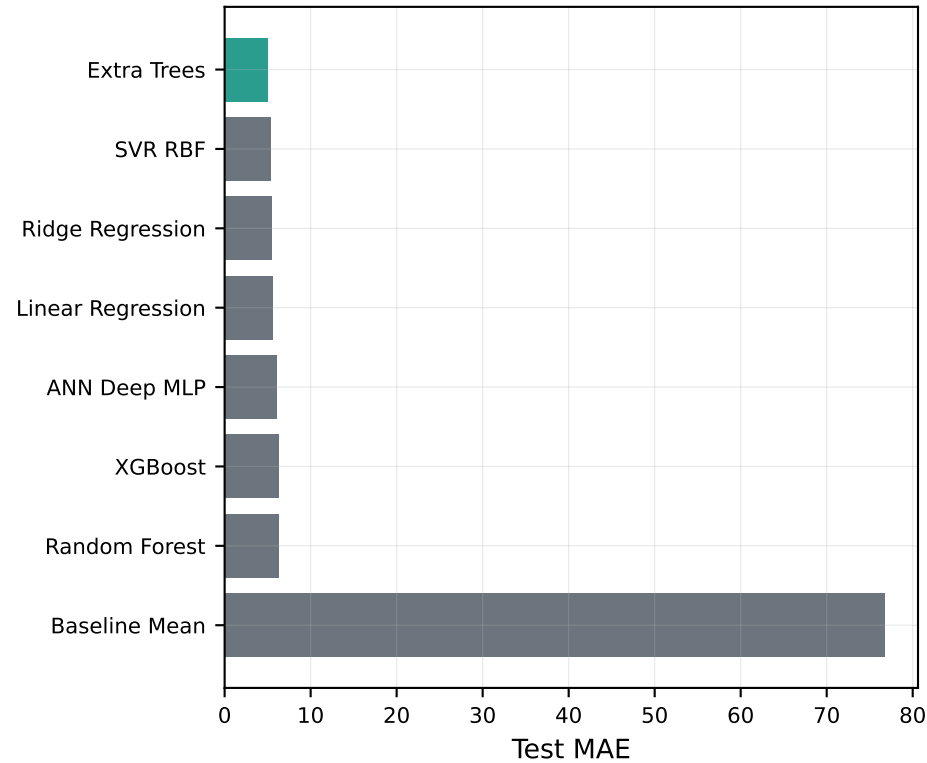


Model Comparison Using Depth-Holdout Test MAE

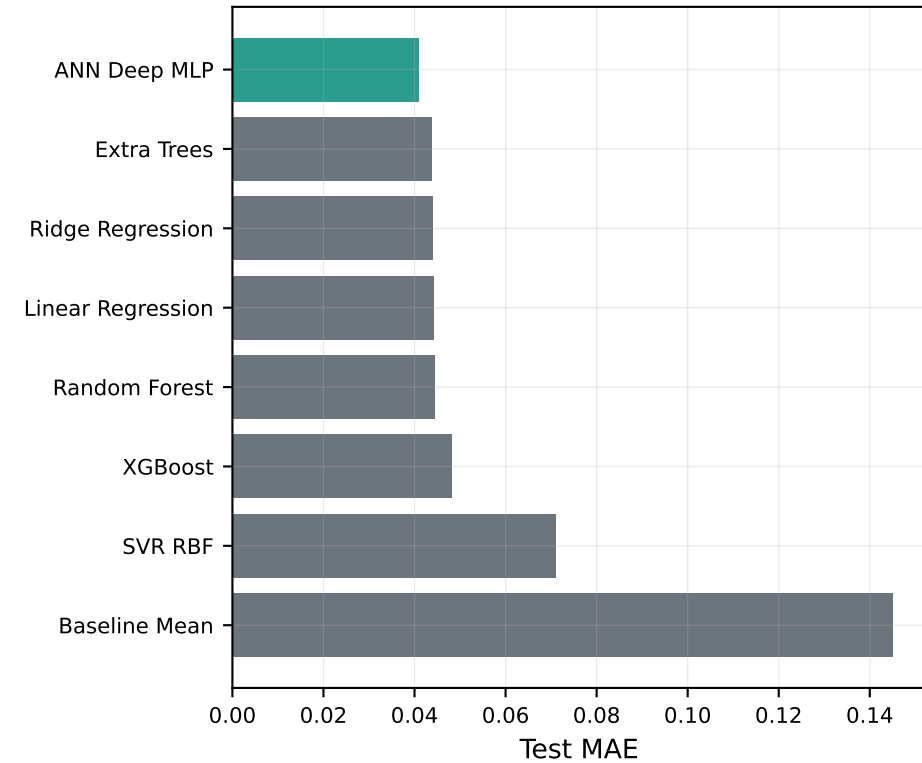
Porosity



Permeability k

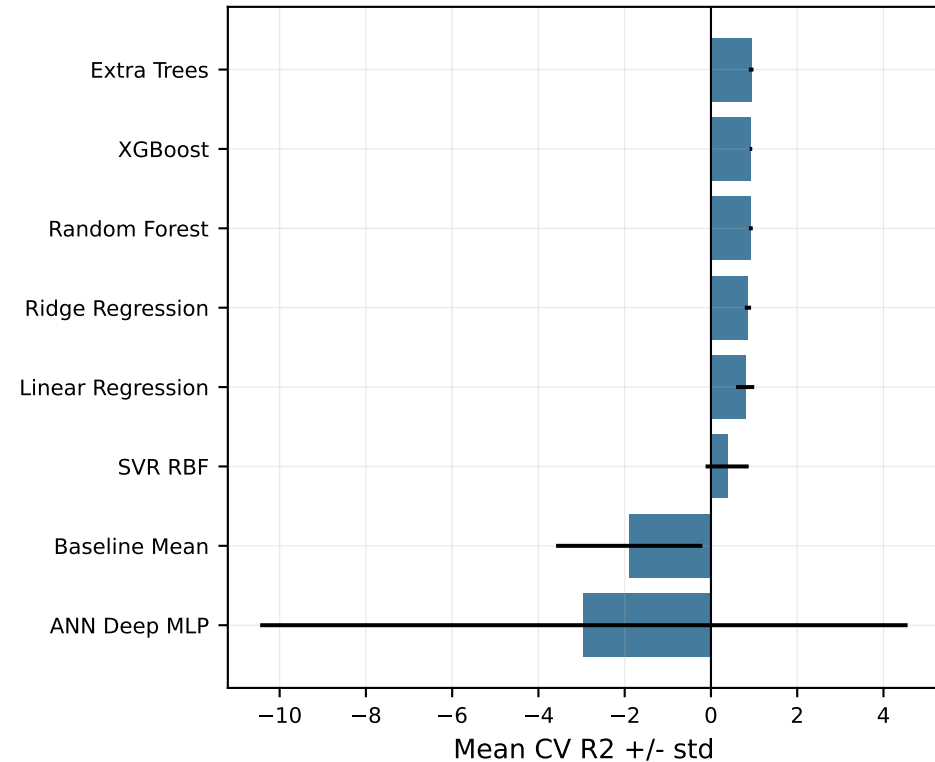


Water Saturation Sw

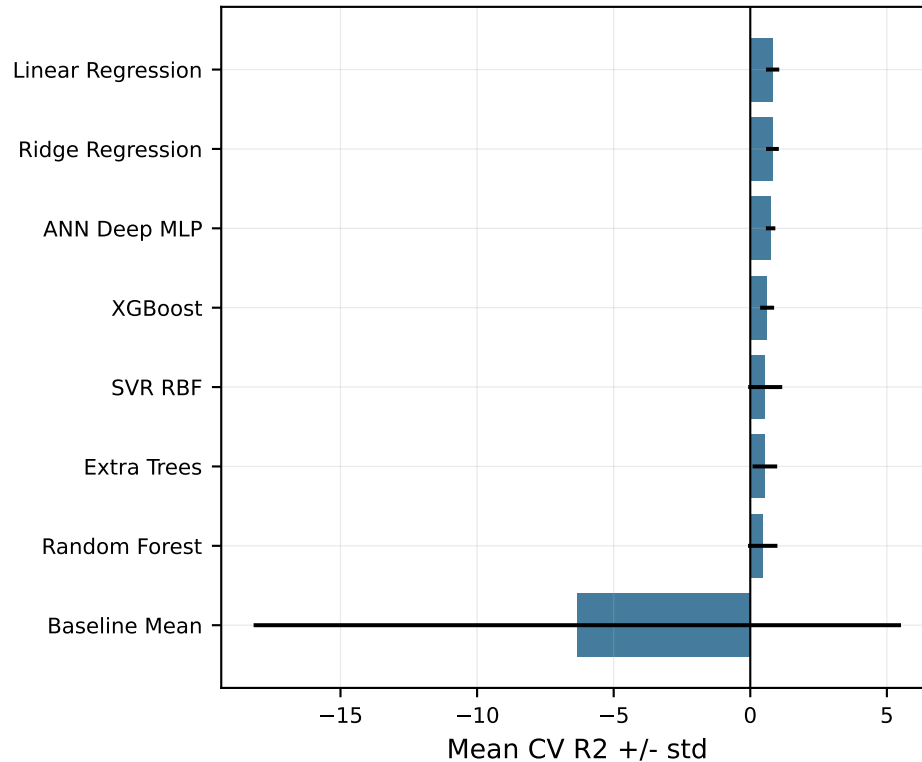


Five-Fold Depth-Ordered Cross-Validation R2

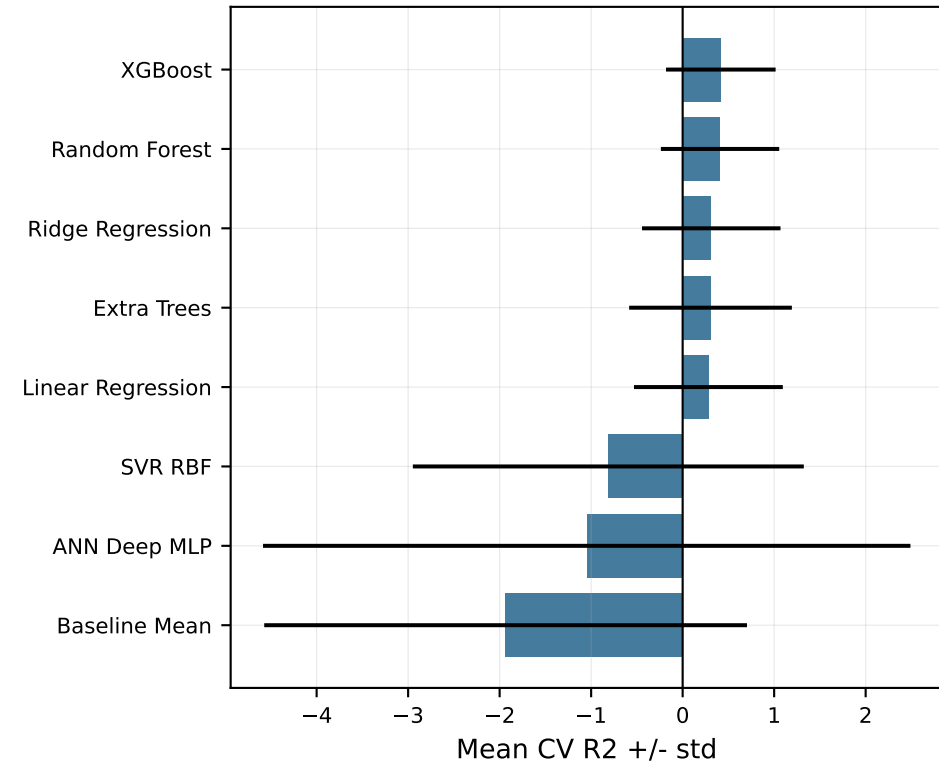
Porosity



Permeability k

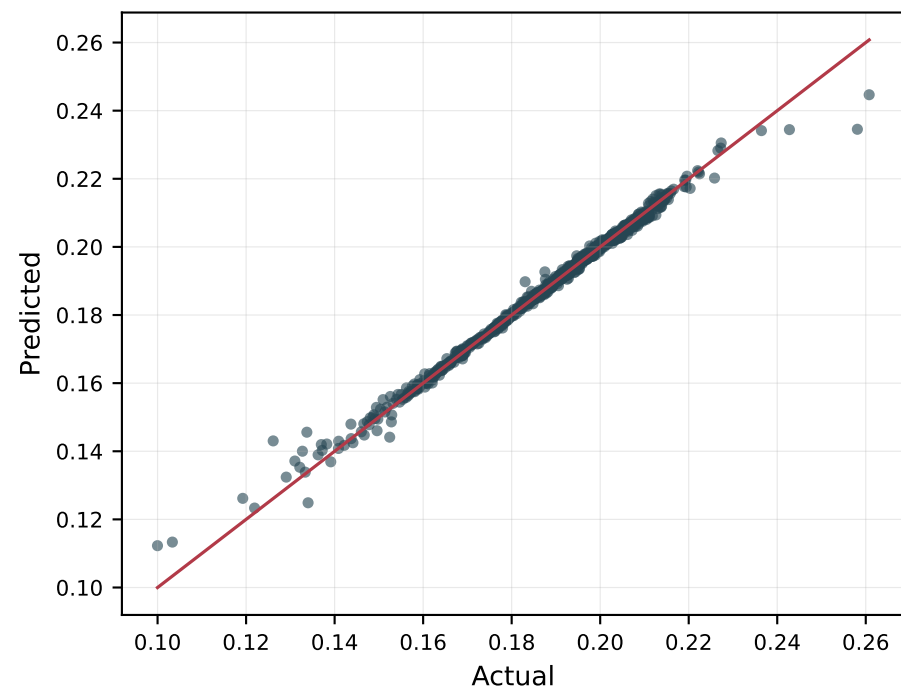


Water Saturation Sw

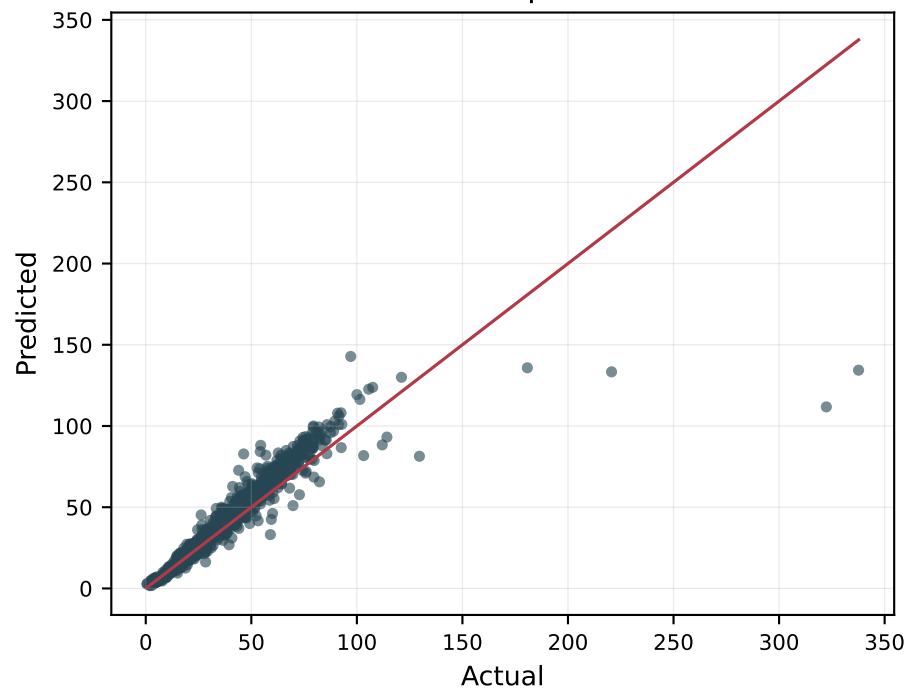


Best Model Predicted vs Actual on Depth-Holdout Test Interval

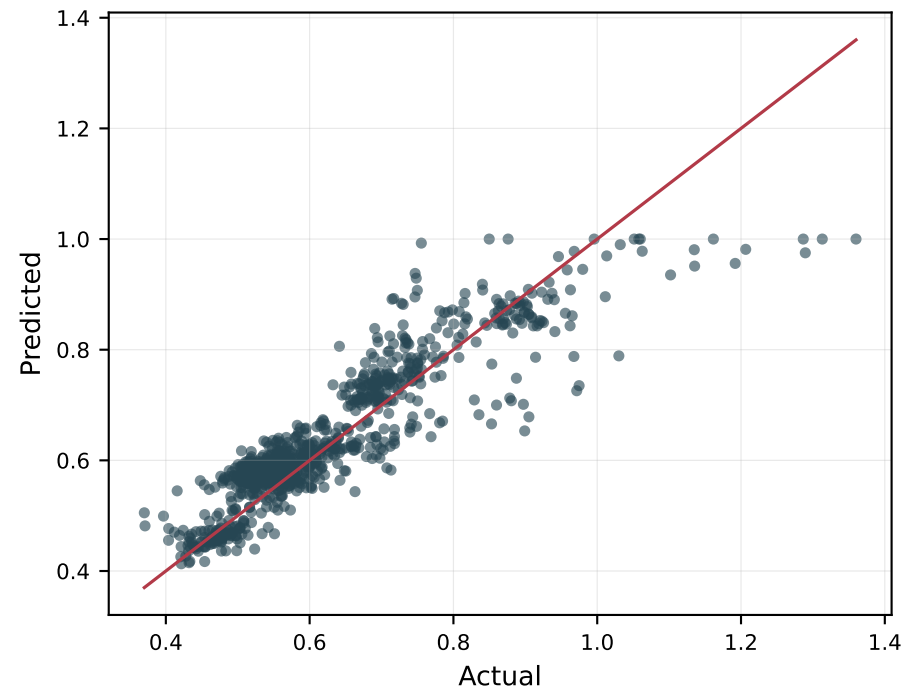
Porosity
Random Forest



Permeability k
ANN Deep MLP

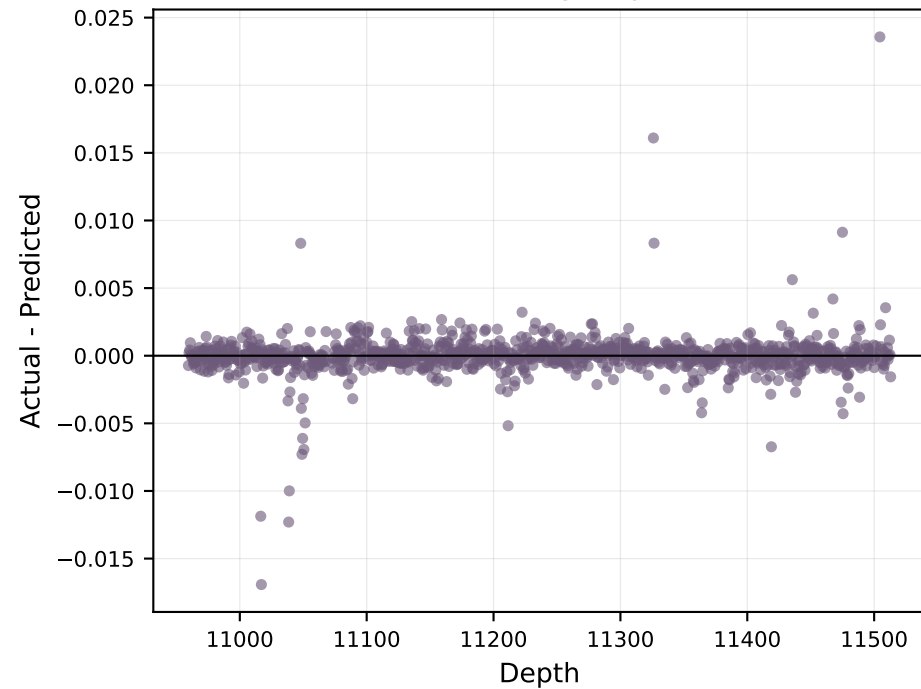


Water Saturation Sw
Random Forest

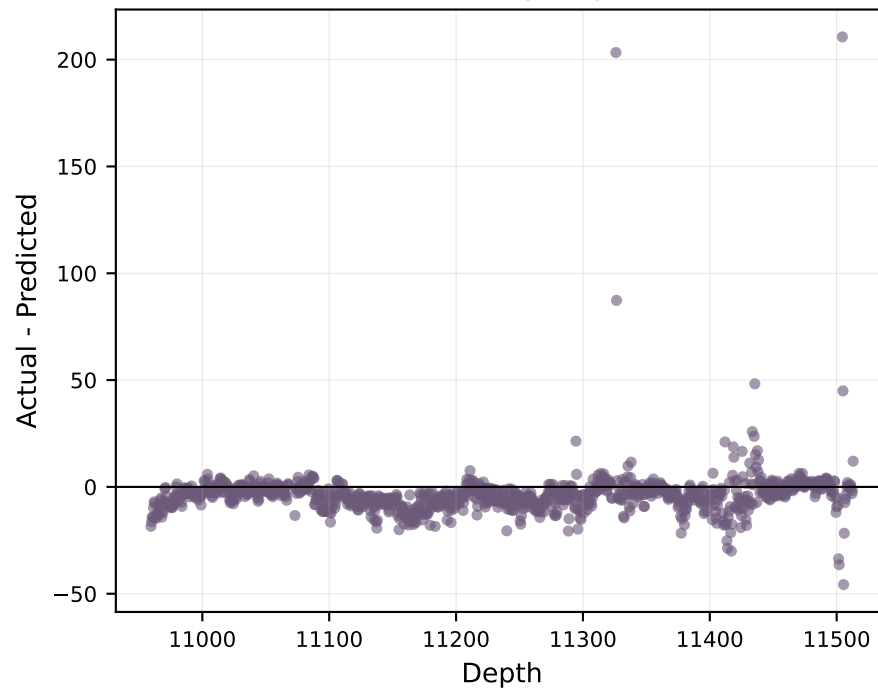


Best Model Residuals Across the Depth-Holdout Test Interval

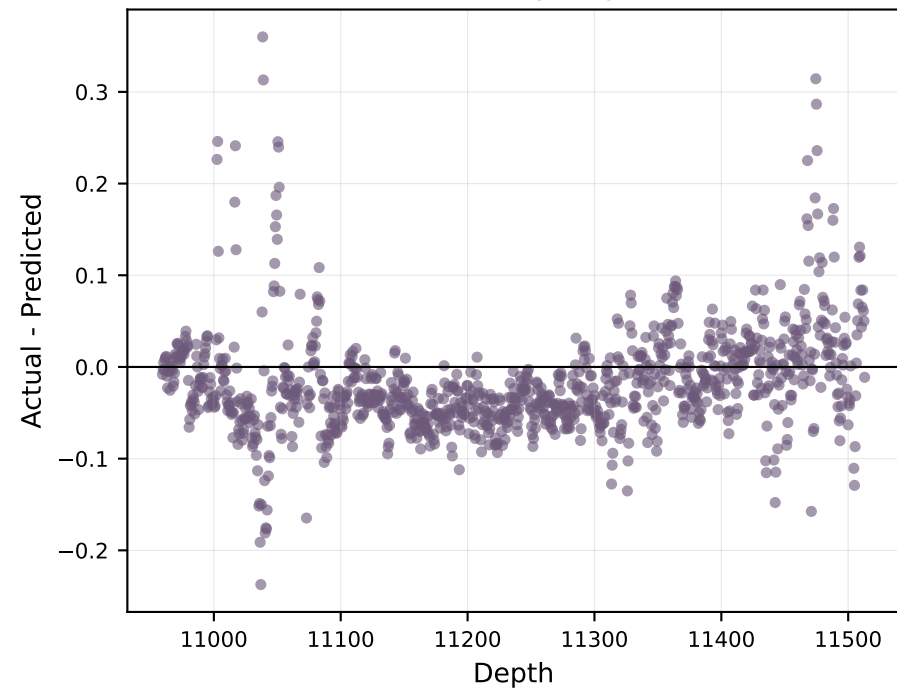
Porosity
Residual by depth



Permeability k
Residual by depth

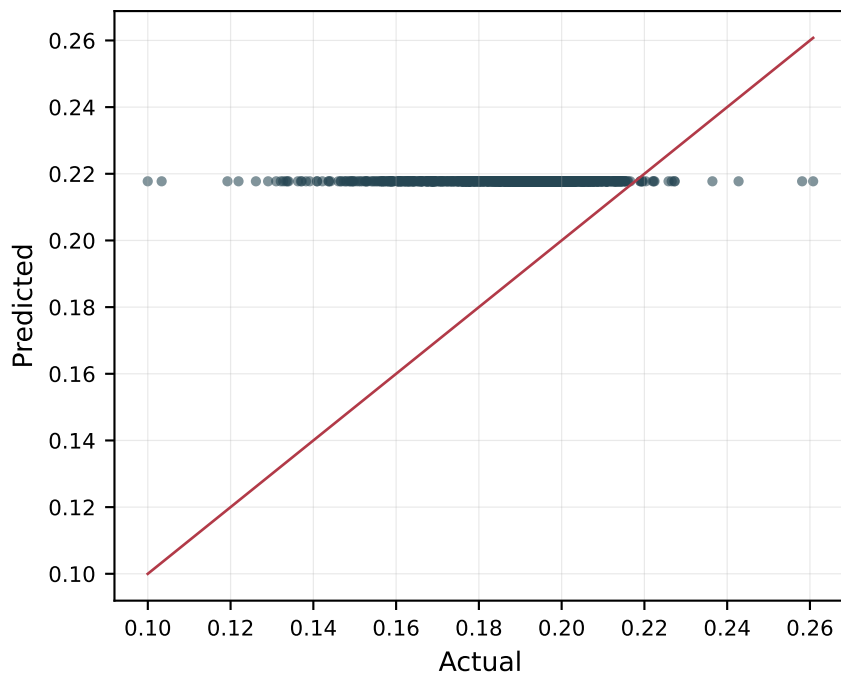


Water Saturation Sw
Residual by depth

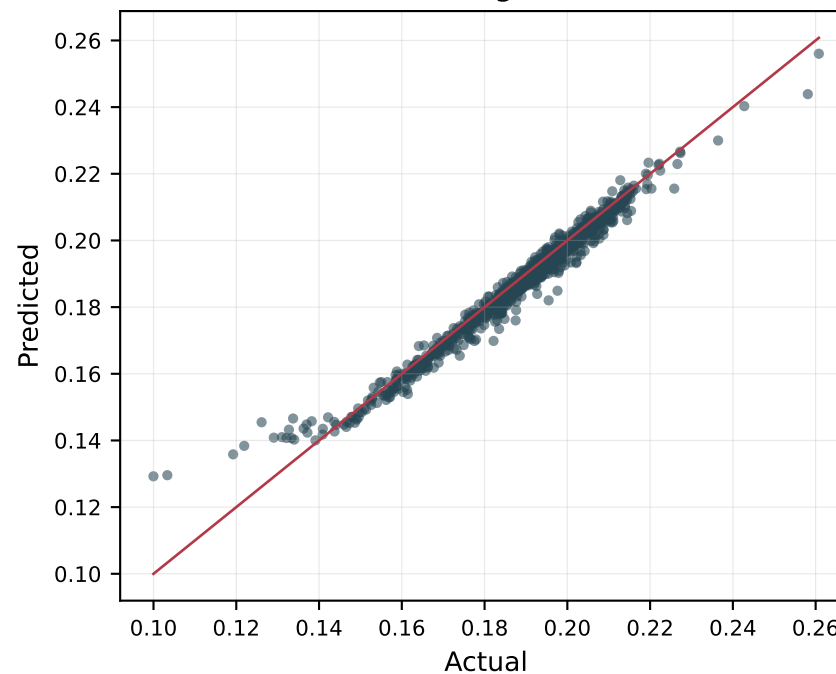


All Models Predicted vs Actual - Porosity

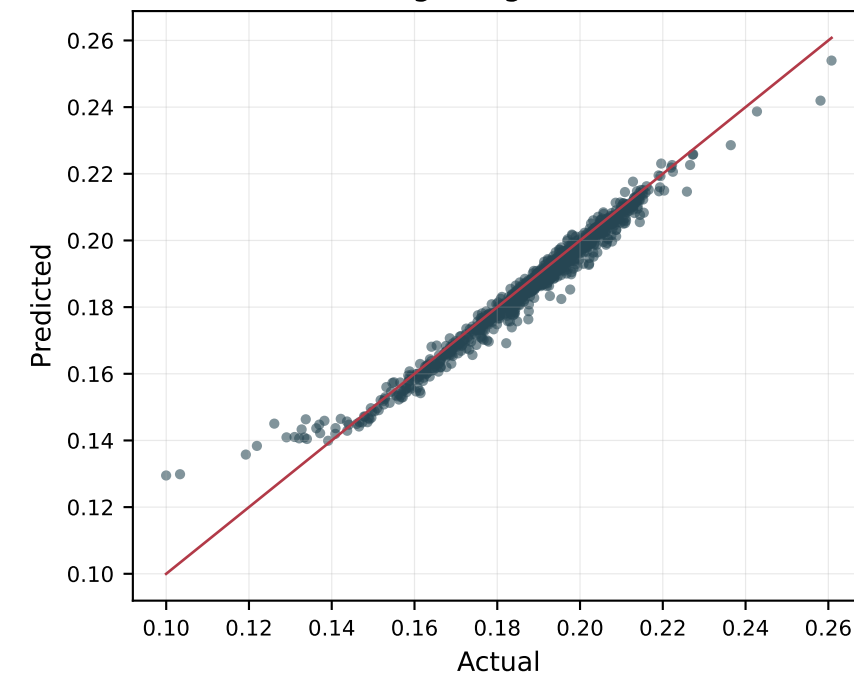
Baseline Mean



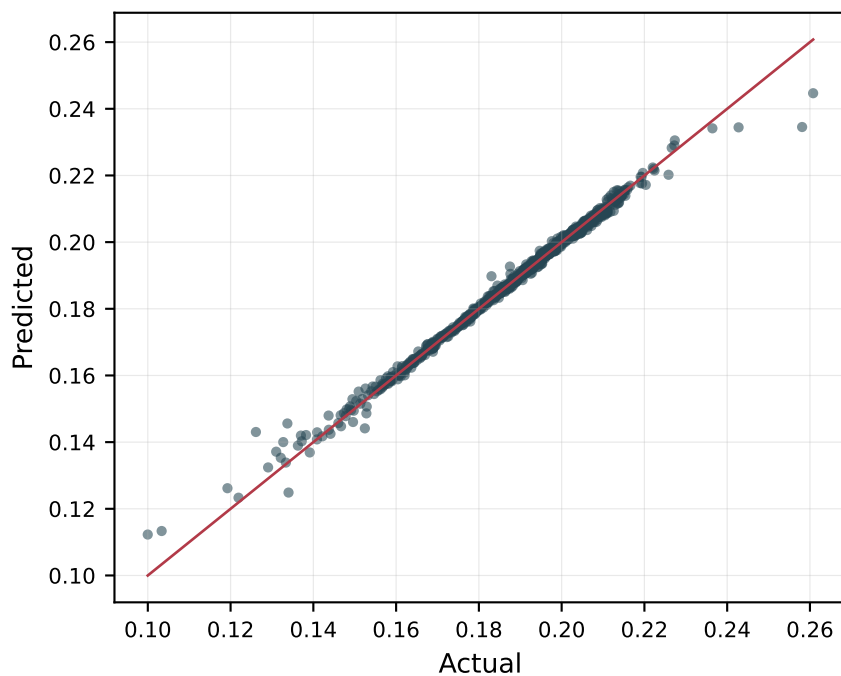
Linear Regression



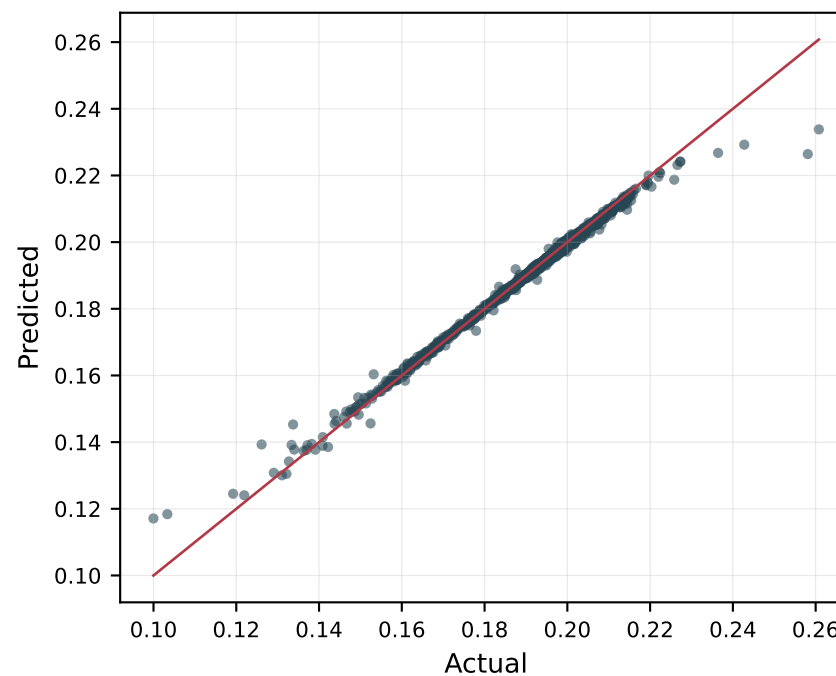
Ridge Regression



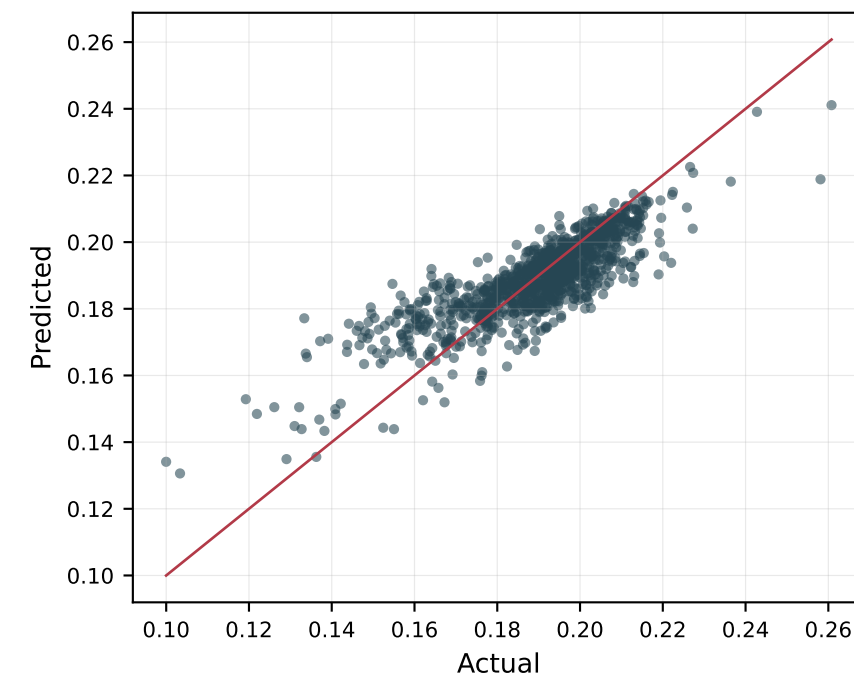
Random Forest



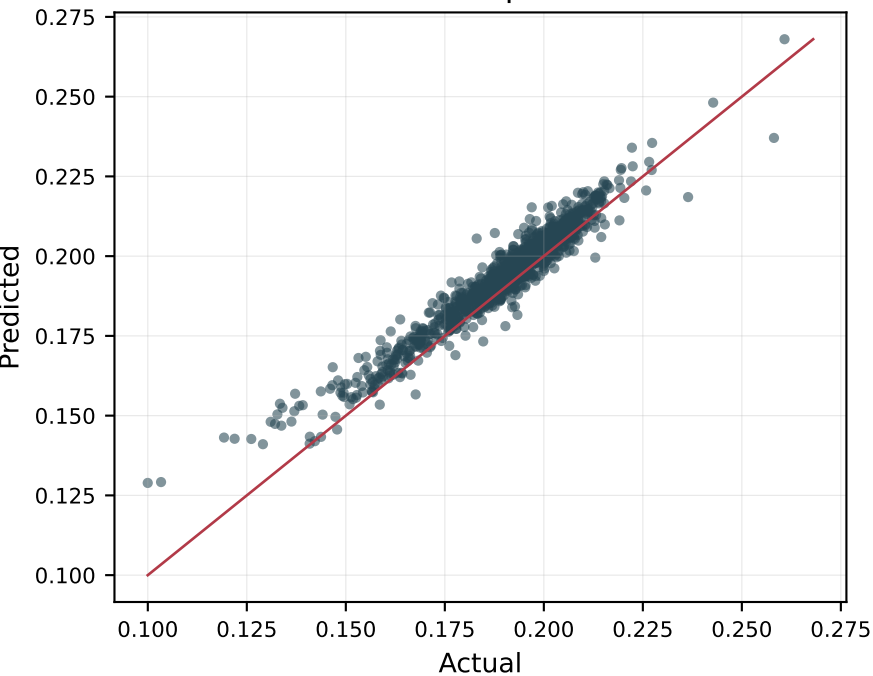
Extra Trees



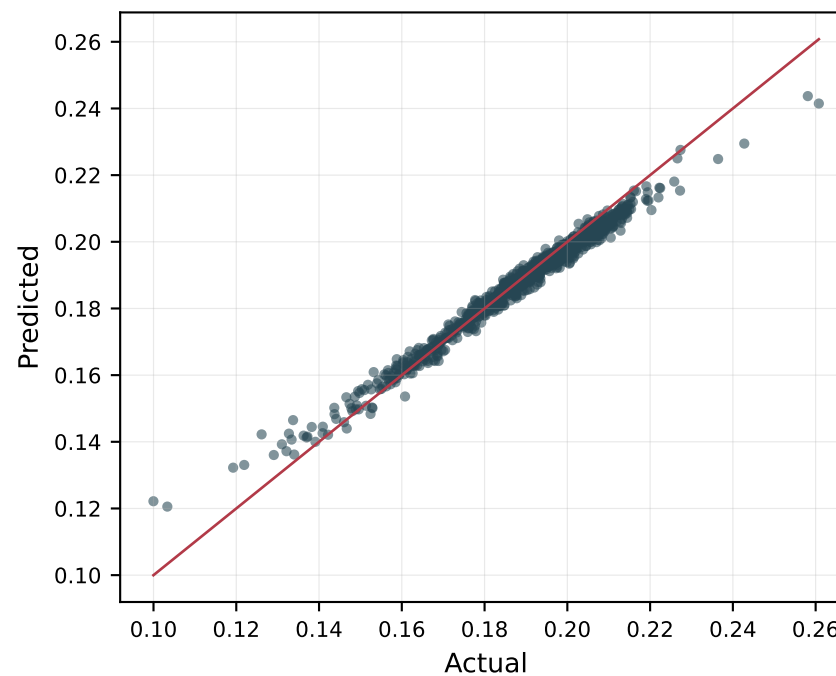
SVR RBF



ANN Deep MLP

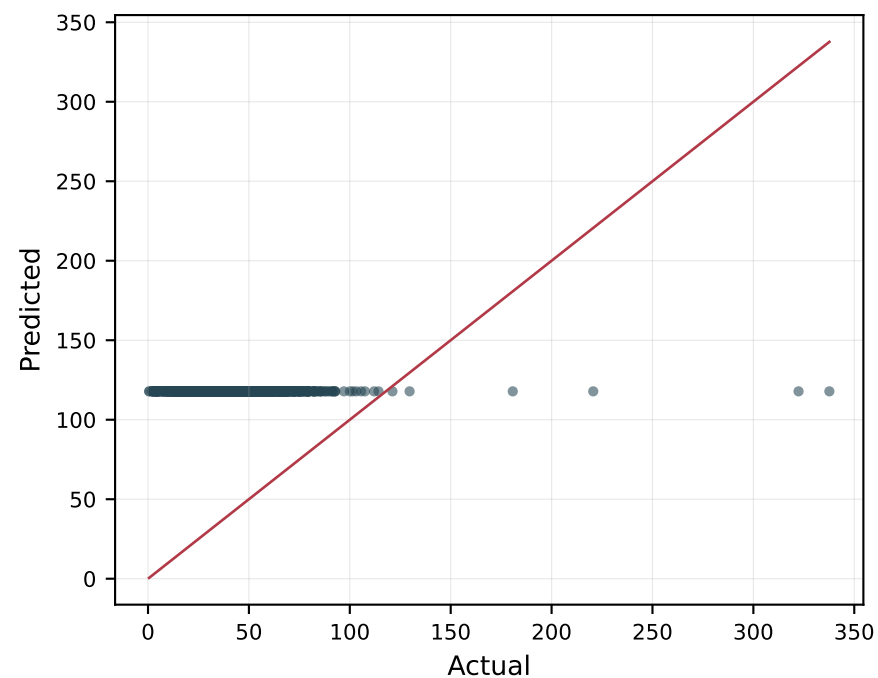


XGBoost

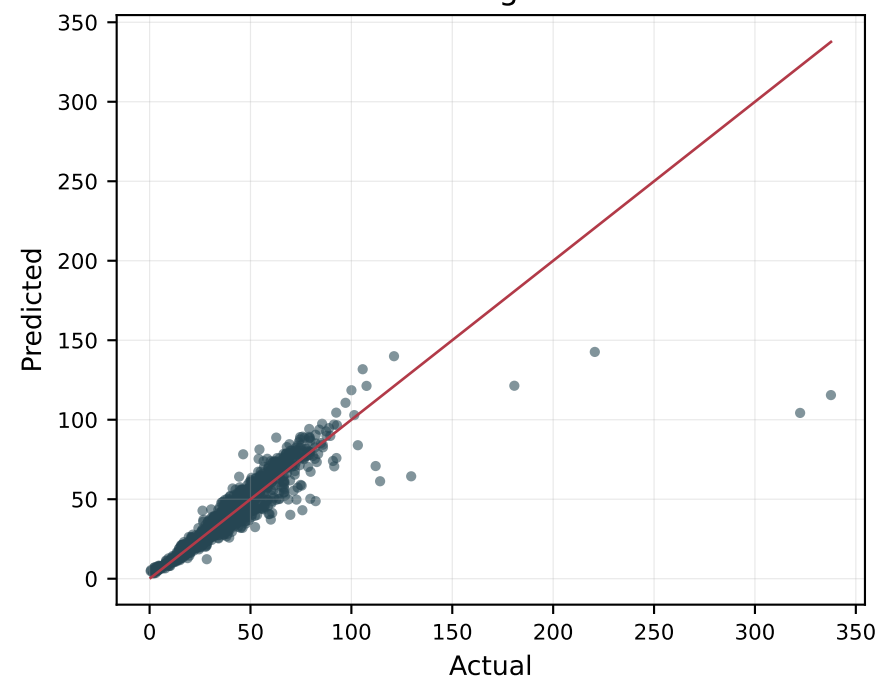


All Models Predicted vs Actual - Permeability k

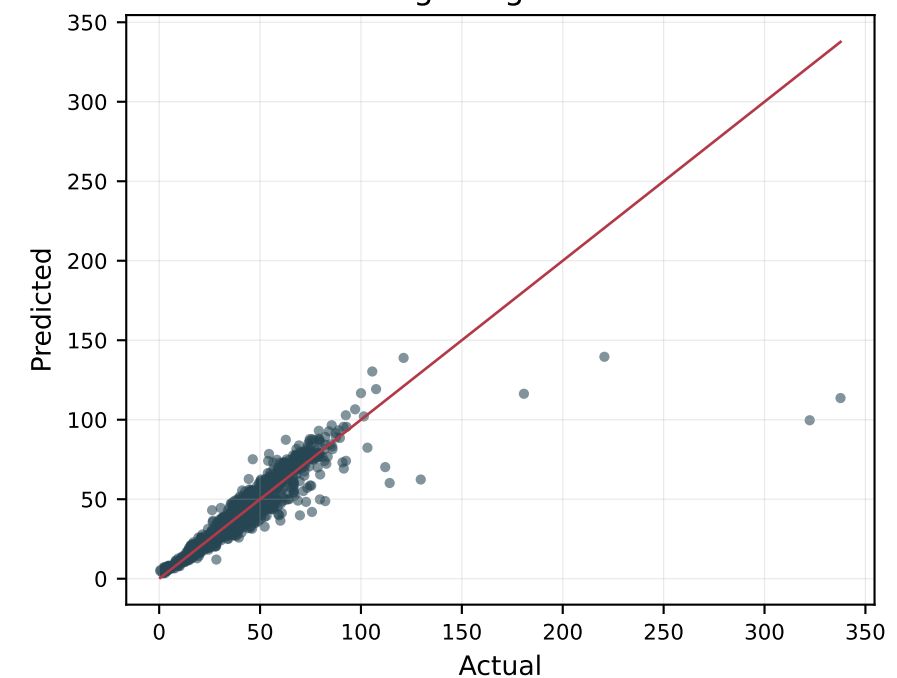
Baseline Mean



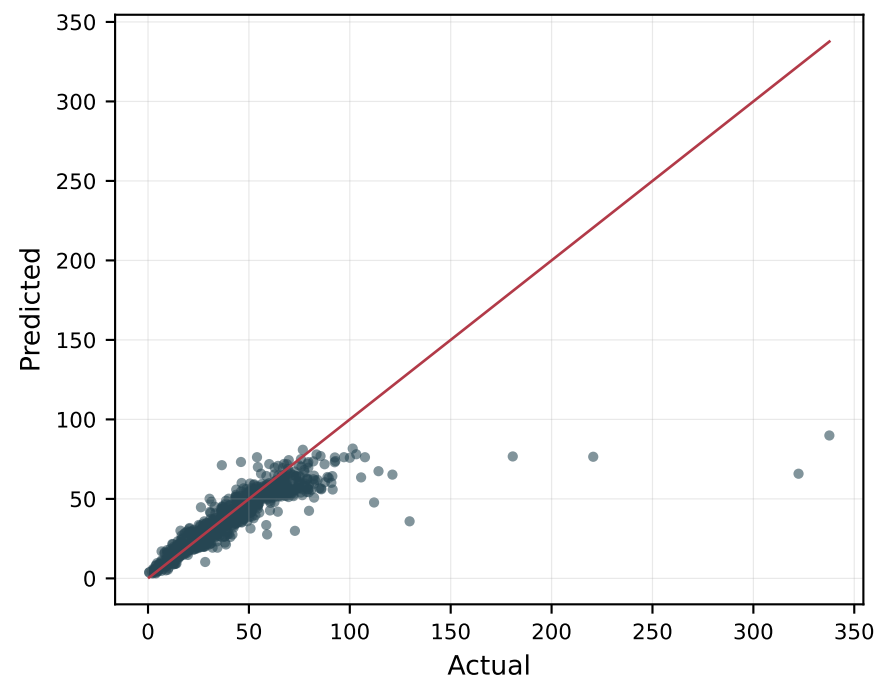
Linear Regression



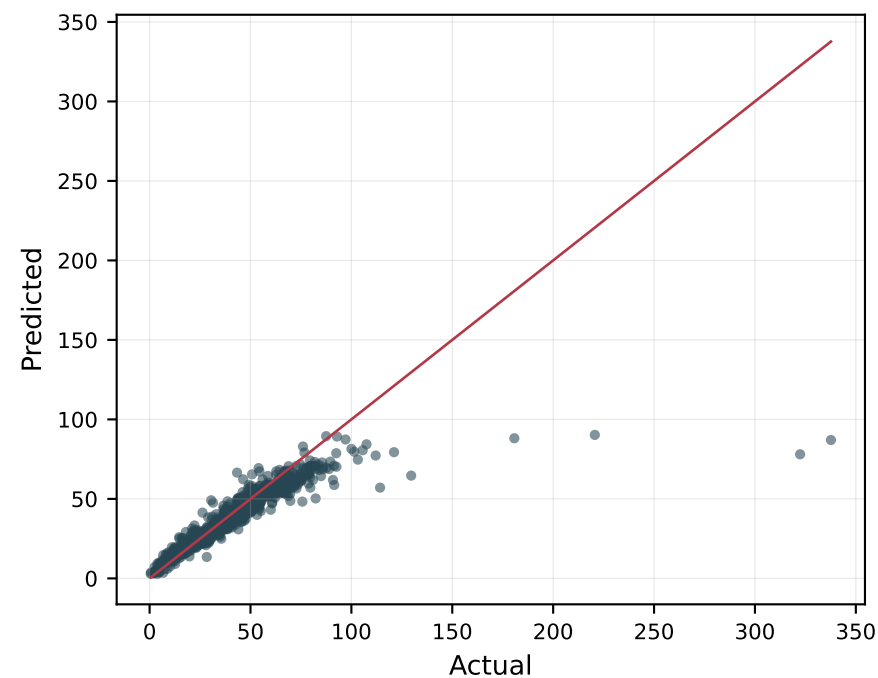
Ridge Regression



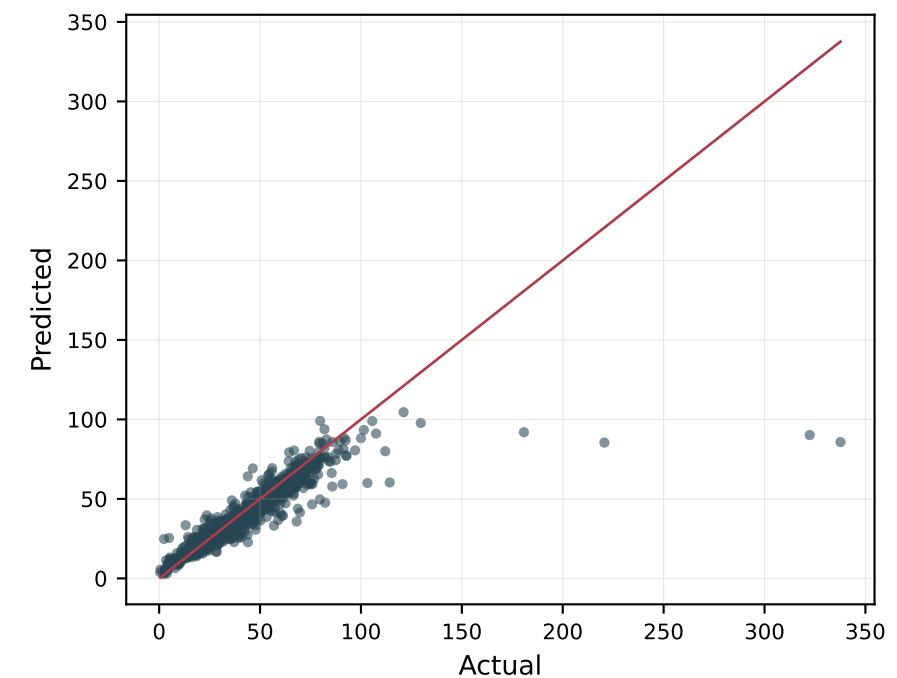
Random Forest



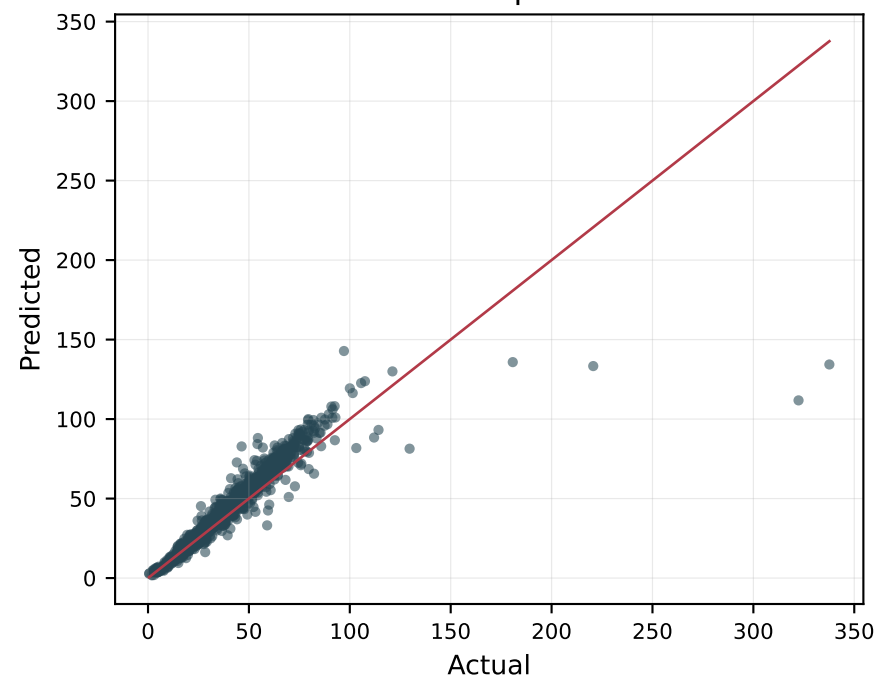
Extra Trees



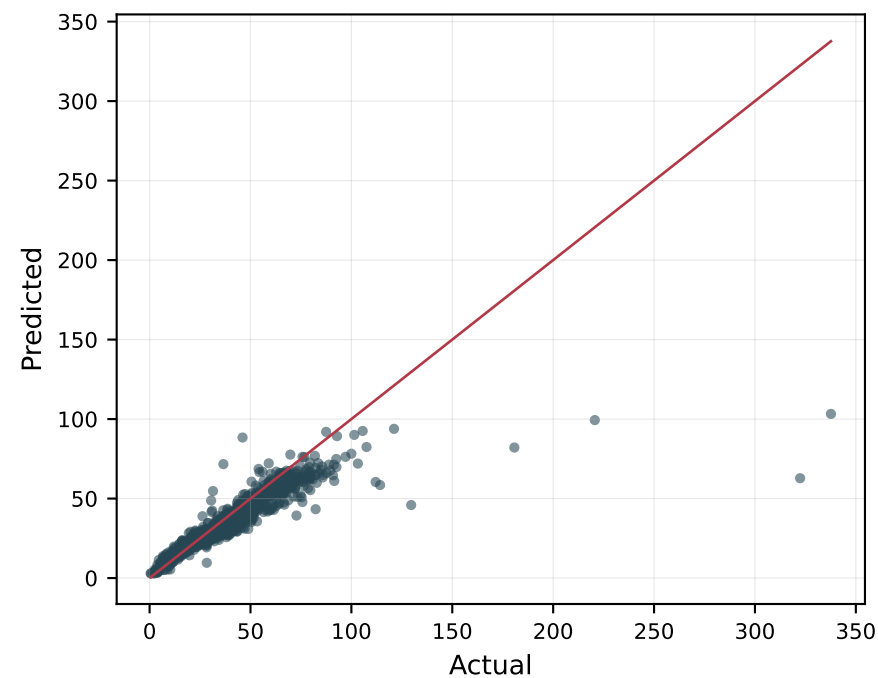
SVR RBF



ANN Deep MLP

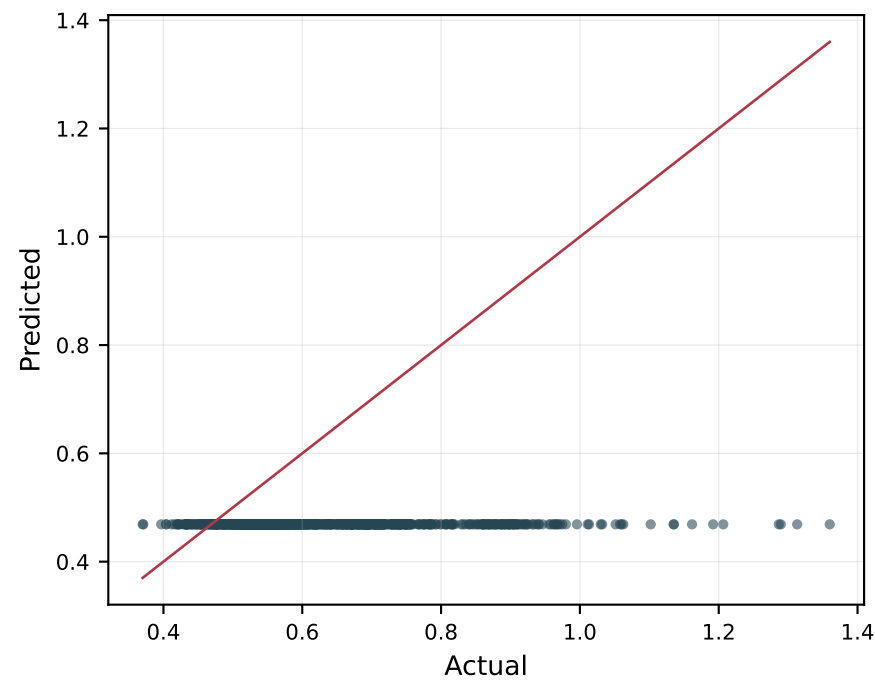


XGBoost

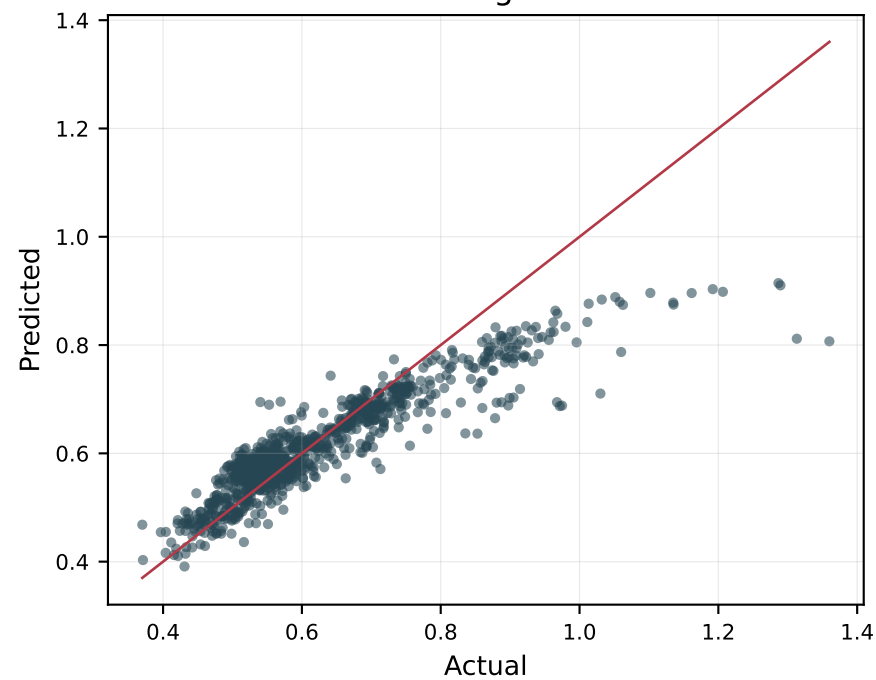


All Models Predicted vs Actual - Water Saturation Sw

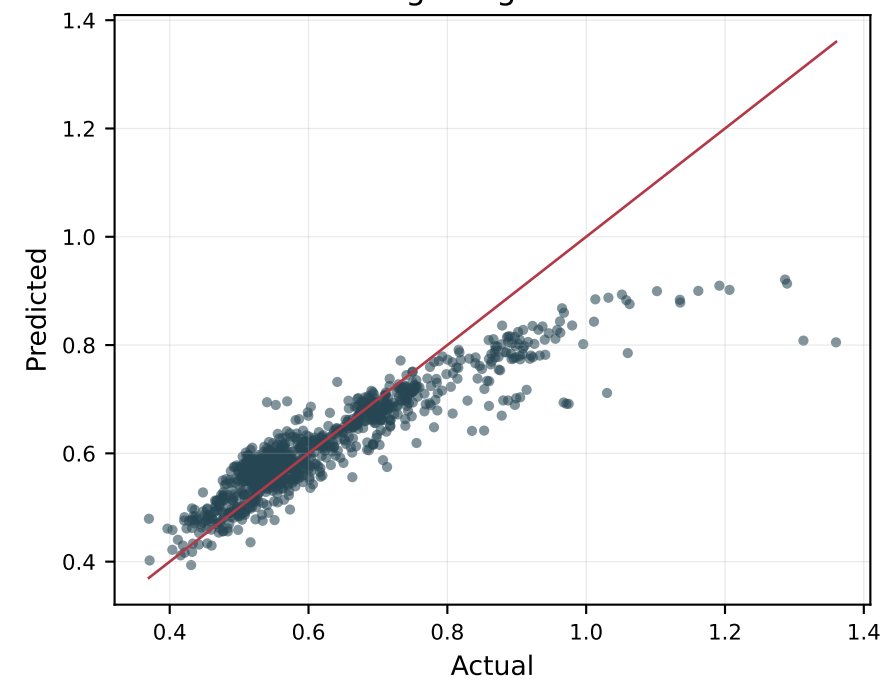
Baseline Mean



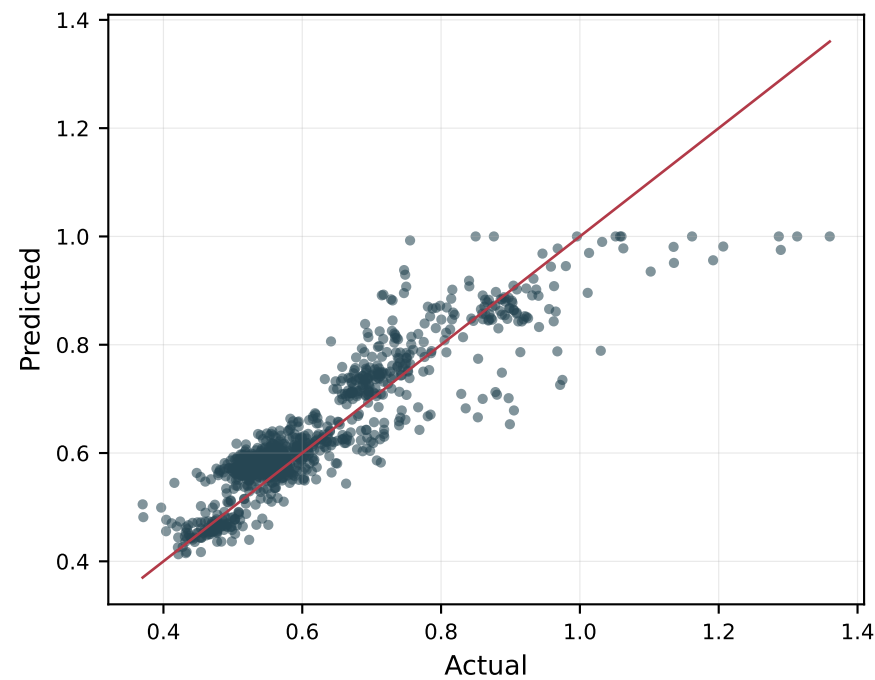
Linear Regression



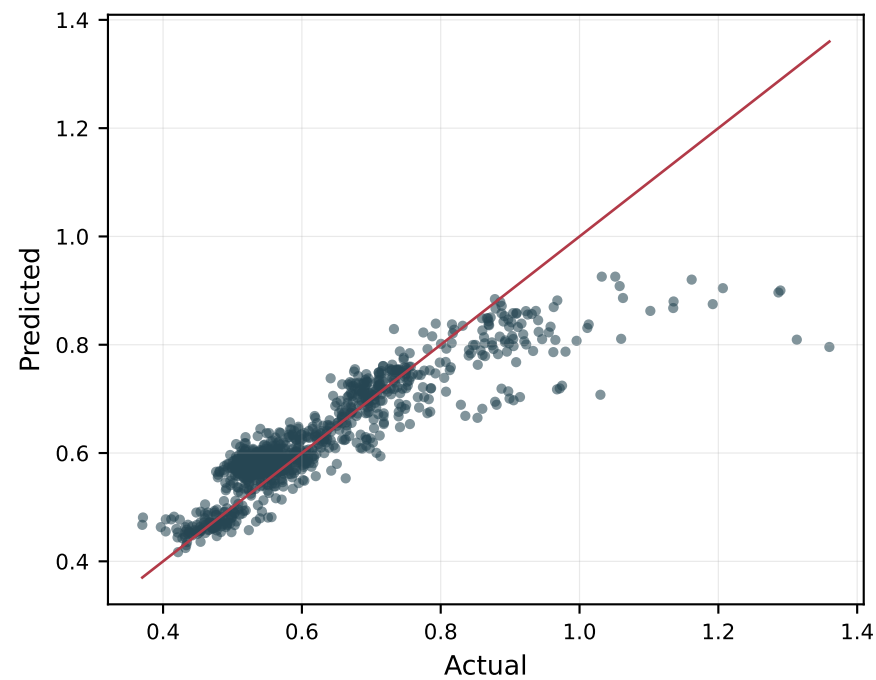
Ridge Regression



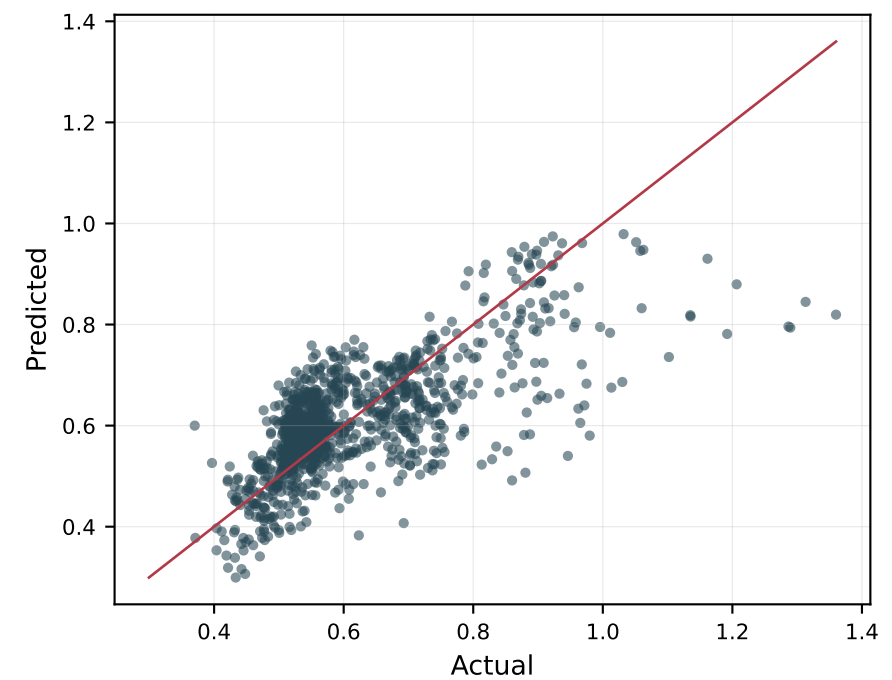
Random Forest



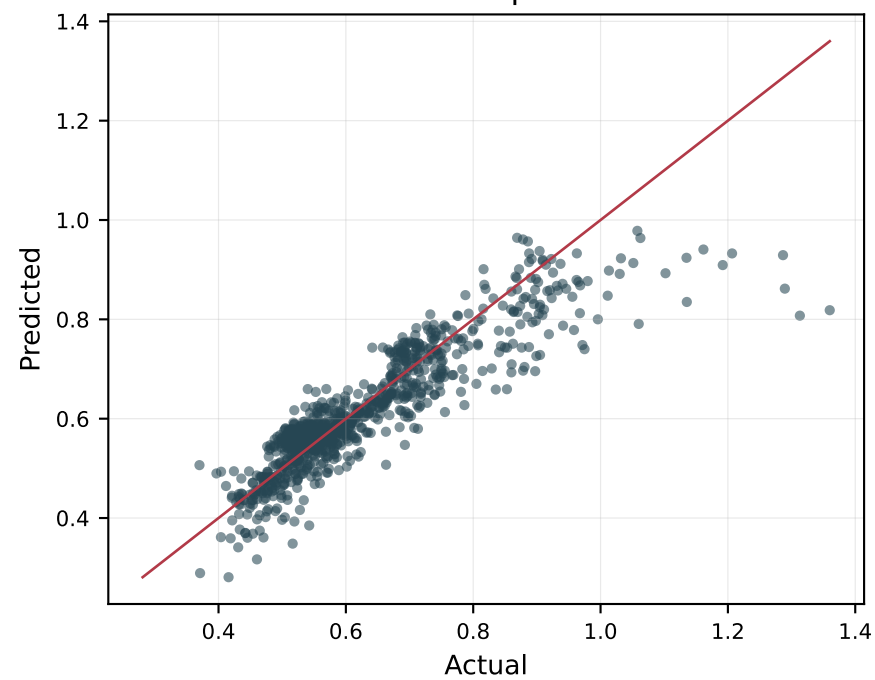
Extra Trees



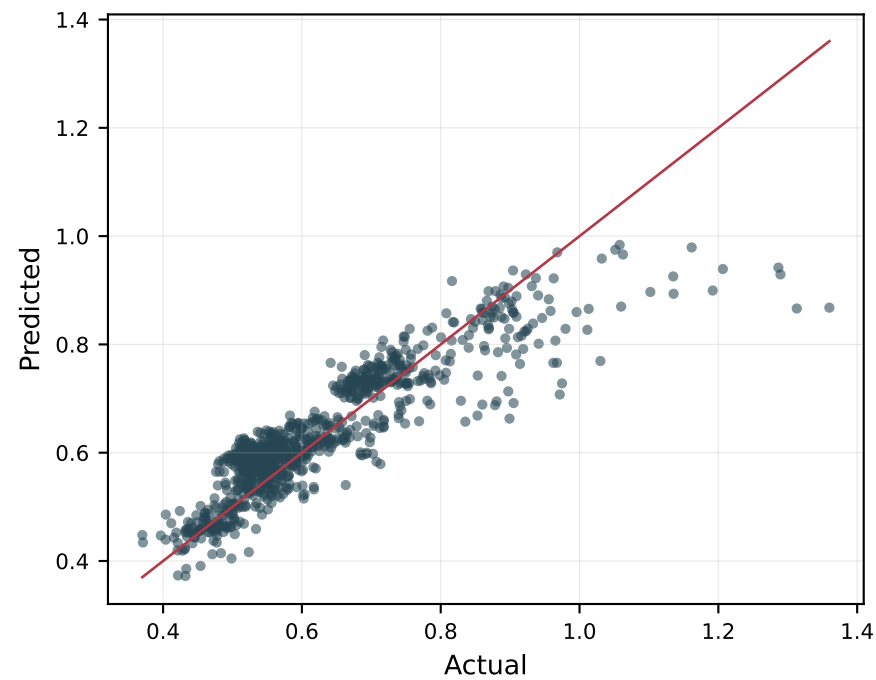
SVR RBF



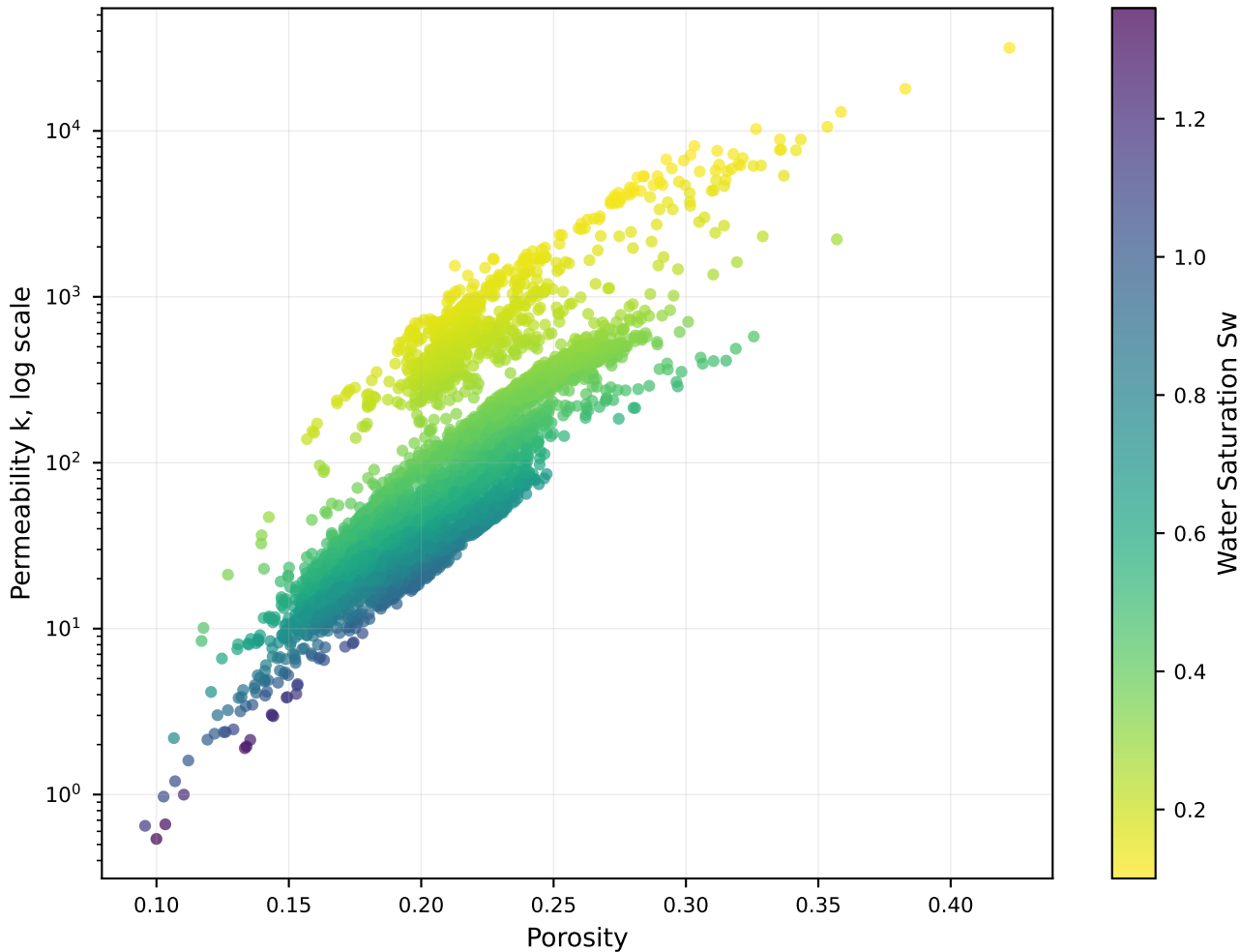
ANN Deep MLP



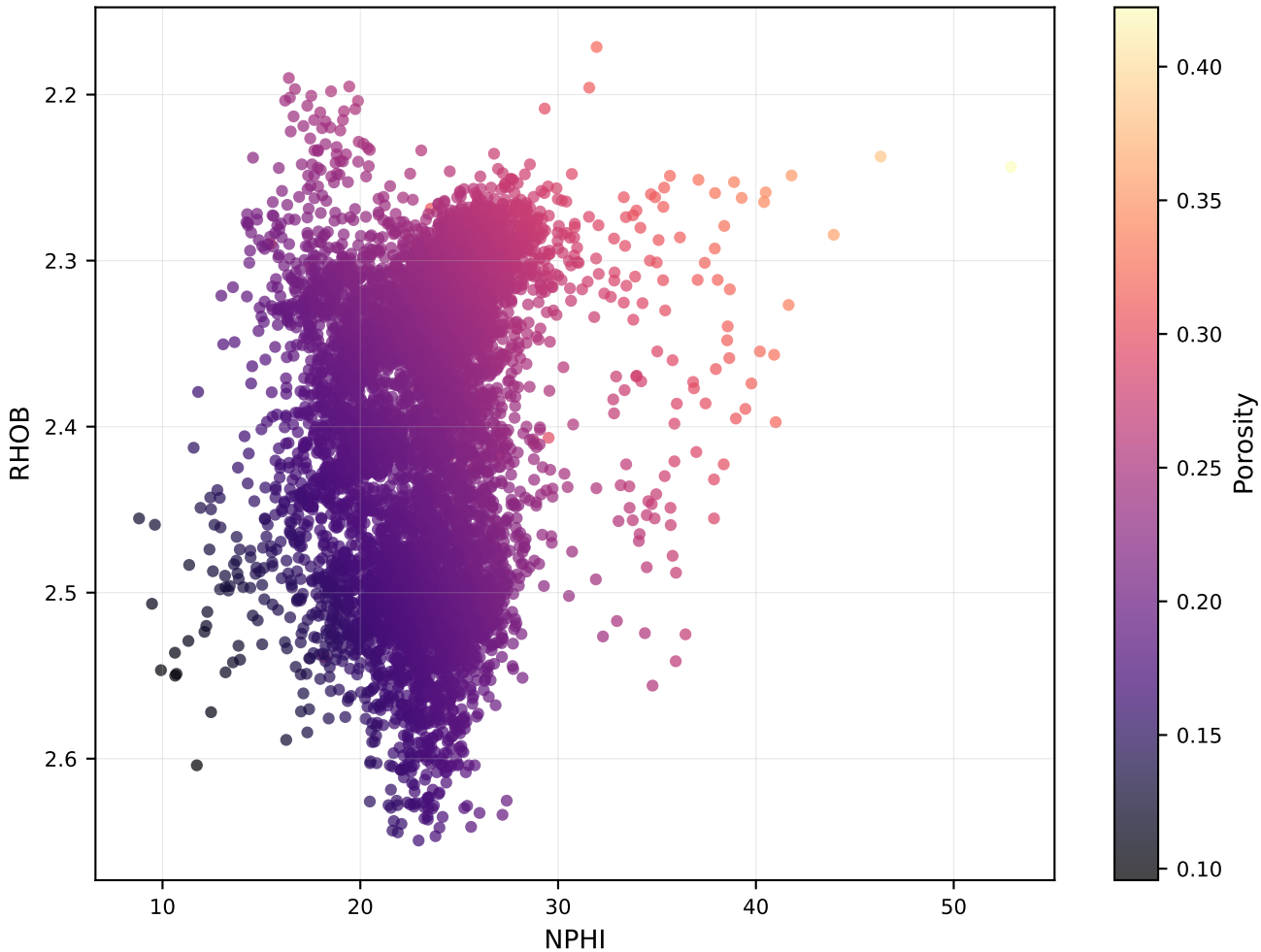
XGBoost



Reservoir Quality Crossplot: Porosity vs Permeability

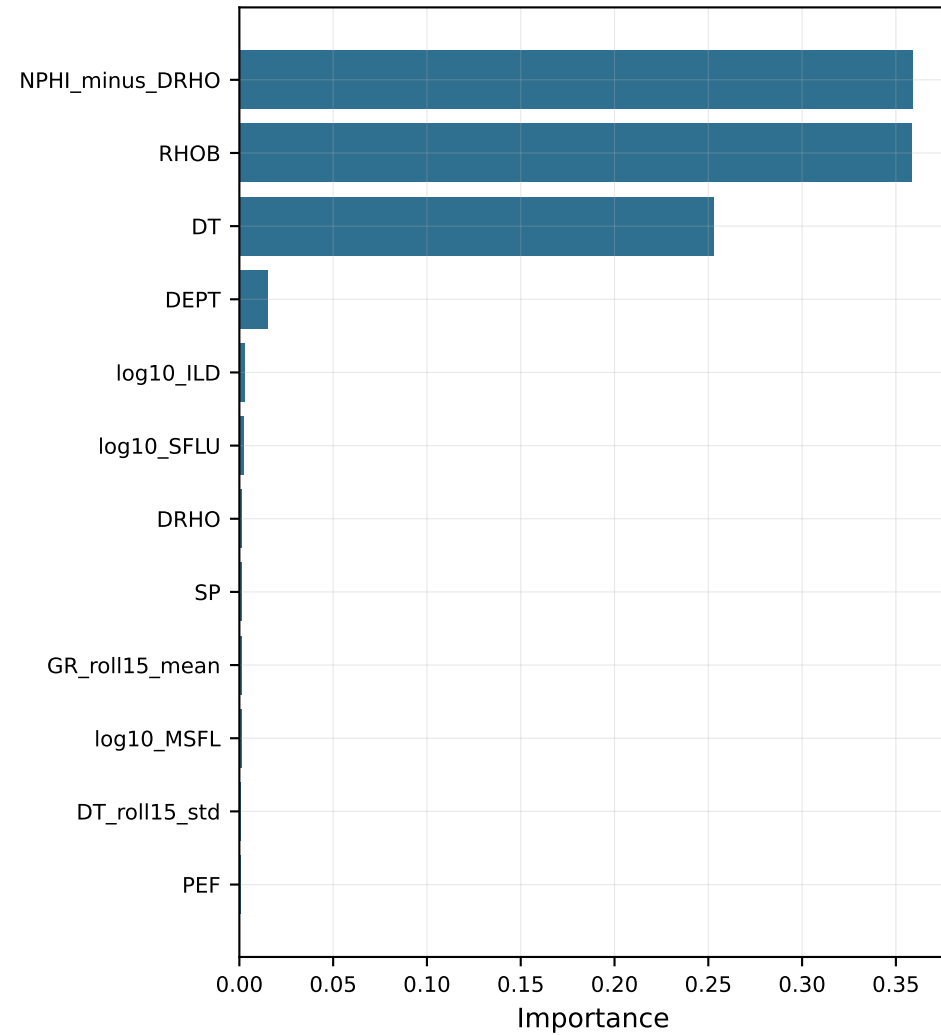


Neutron-Density Crossplot Colored by Porosity



Most Important Inputs for Best Models

Porosity



Permeability k

Water Saturation Sw

